CNO-EVIL

Pilot Manual

How every mechanic of the mod works, what you see in the cockpit and what to do about it. Version 1.1.5, for Nuclear Option.

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Part I. Getting started

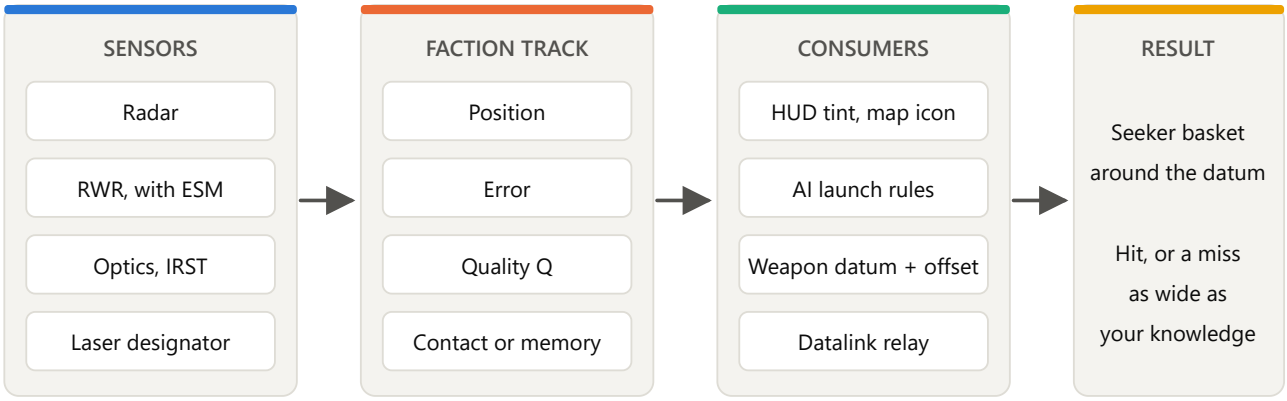
1. What this mod does

In vanilla Nuclear Option, knowledge is almost free. Once something is known, every unit of your faction sees it at its true position, every RWR points exactly at every radar, and missiles fly to the real target.

CNO-EVIL replaces that with one rule:

> **Your faction knows only what its sensors actually find, and your weapons fly to what your faction knows.**

Everything else in this manual follows from that sentence. A radar is now a physical machine that can fail to see you. A warning receiver gives you a direction with an error, not a position. A missile fired at a poor track flies to the wrong place. The AI lives by the same rules, so an enemy that has not found you really has not found you.



From sensor to weapon: every chapter of this manual sits somewhere on this chain.

What changes, at a glance

Area	Vanilla	With CNO-EVIL	Chapter
Radar detection	Signal falls with distance	Radar equation: power, RCS, noise, clutter, jamming, Earth curvature, terrain, sea reflection	4
Radar cross section	One number per aircraft	Changes with aspect, stores, bay doors, rotor blades	5
Ground clutter	Small signal penalty	Doppler notch: fly across a look-down radar and it can lose you	6
RWR	Exact direction to every radar	Direction with an error; quality depends on your aircraft (Tier 1 to 4)	7
Enemy radar positions	Known when your RWR hears them	Known only after a fix, a radar contact or a visual contact	12

Area	Vanilla	With CNO-EVIL	Chapter
Knowledge	True positions, shared by all	Tracks with a quality Q; the track, not the truth, drives displays and weapons	10, 11
Jamming	Simple jam value	Jammer power against echo power, burn-through, home-on-jam, a shared power budget	14, 15
Countermeasures	Flares and jammer	Budget of flares, chaff and active decoys by aircraft (at least 1.4 x the vanilla flares; fighter about 90, Medusa 127, bomber up to 192), chosen in the loadout menu; two one-key programs	16 to 18
Weapon accuracy	Fly to the true target	Fly to the faction estimate plus an offset that depends on Q	19
Active radar missiles	Warning from the missile's target data	Seeker is silent until terminal range; only real sensors warn you	20, 24
AI	Fires when the target is known	Fires when track quality and hit chance are good enough; radars go silent against an incoming ARM	25
Air defense positions	Fixed by the mission file	New layered sites at each mission start	25, 31
HUD and map	All known units at true positions	Markers tinted by Q; map icons only for located threats	26, 28
TacScreen	Vanilla MFD	PortalMFD: seven pages on two or more screens, by keyboard	29

How to read this manual

Each mechanic chapter has the same shape, so you can jump straight to what you need:

- **Vanilla** and **With the mod**: what changed.
- **How it works**: the model in plain words.
- **What you see**: where it shows up in the cockpit.
- **How to use it**: what to do about it.
- **Common mistakes**: what catches new pilots.
- **Numbers** and **config keys**: default values, for the curious and for tuning.

All numbers are the defaults of version 1.1.5. A host or a player can change them (chapter 37). This manual describes the model inside the game, nothing else.

2. Install, keys and the F10 switch

Install

You need Nuclear Option and BepInEx 5 for Unity Mono, x64.

1. Close the game.
2. Create the folder `BepInEx\plugins\CNO-EVIL` and put `CNOEVIL.dll` in it.
3. Start the game, then open `BepInEx\LogOutput.log`.

4. Look for `CNO-EVIL Harmony patches applied: N succeeded, 0 failed`. If the failed number is not 0, report it.

The first start creates the settings file `BepInEx\config\com.cnoevil.nuclearoption.overhaul.cfg`. Do not edit it while the game runs.

Keys

Key	What it does
`F10`	Switch between CNO-EVIL and vanilla, live
`;`	PortalMFD: next page on the left screen (Left Shift held: next range)
`'`	PortalMFD: next page on the right screen (Left Shift held: next range)
`/`	PortalMFD: next page on the centre screen, on cockpits that have one (Left Shift held: next range)
`.`	PortalMFD: next page on the lower centre screen, on cockpits that have one (Left Shift held: next range)
`Right Shift` (tap)	Small cockpit screen: attitude page or datalink page
`LeftAlt` (tap)	Countermeasure program 1: one flare, one chaff, one decoy
`C`	Countermeasure program 2: two flares, two chaff
`B`	Open or close the HMD thermal window
`V`	Thermal window zoom: 1x, 5x, 10x (set in `HMDThermalZoomLevels`)
`]` / `[`	Cockpit minimap zoom in / out
`F9` / `F8` / `F11`	Debug overlays: telemetry, datalink, radar rays
`F6`	Log mark: writes a numbered mark and a snapshot to the session log (needs `DebugLogging`, chapter 38)

The range modifier is `PortalMFDRangeModifierKey` (Left Shift). Set it to None and the page keys change only the page.

Vanilla controls keep working. The vanilla countermeasure button fires the selected station, and Next Countermeasure now cycles flares, jammer, chaff and decoy. A tap of `LeftAlt` or `Right Shift` means 0.5 s or less with no other key, so Alt+Tab never dumps your chaff, and Shift with a page key does not change the small screen.

The debug overlays show things a pilot would not know. Use them to learn, not to fight. In multiplayer, F8 to F11 work only in single player and on the host. F6 works on clients too.

F10: your best teacher

F10 switches the whole mod off and on in flight. If something surprises you, press it and compare. On a multiplayer client the key does nothing: the client follows the host.

3. Your first flight

Take an FS-12 Revoker, a simple Tier 3 aircraft with a radar, and fly a mission with one or two enemy SAM sites.

1. **In the loadout menu**, find the three new sliders under fuel: `Flares`, `Chaff`, `DRFM`. Leave them at the default: on the Revoker 64, 20, 6 (budget 90, DRFM at most 13).
2. **After takeoff**, look at the right of the HUD. Below the altitude you see `ID NO`: nobody has a track on you. Below that, `RCS` shows your frontal radar cross section once the gear is locked up.
3. **Look at the TacScreen.** Left is the TSD page, right is the RWR page. Press `;` and `'` to walk through the pages.
4. **Fly toward the enemy.** A yellow strobe appears on the map. That is a bearing, not a position. The radar is somewhere in the circle at the end of the cone, and not at its centre.
5. **Turn about 90 degrees** to that bearing and hold the heading. After about 25 seconds the circle shrinks, and after about a minute the radar icon appears on the map: you have located it.
6. **Watch the ID line.** `ID LOW`, then `ID MED`: the enemy is building a track on you. Watch the small bar under the max-G readout: when the pip passes the centre tick, a radar sees you.
7. **When the HUD shows** `[ MISSILE ]`, put the radar at your 3 or 9 o'clock, descend so the ground is behind you, and tap `LeftAlt`. Keep turning.
8. **Press F10** once on the way home and see how much the vanilla game was telling you for free.

That one flight touches most of this manual. The next parts explain each step.

Part II. Sensors

4. Radar detection and the horizon

Vanilla

The signal is roughly `maxRange / distance`, scaled by RCS. Past the limit you are invisible, inside it you are seen. Ships are not radar targets at all.

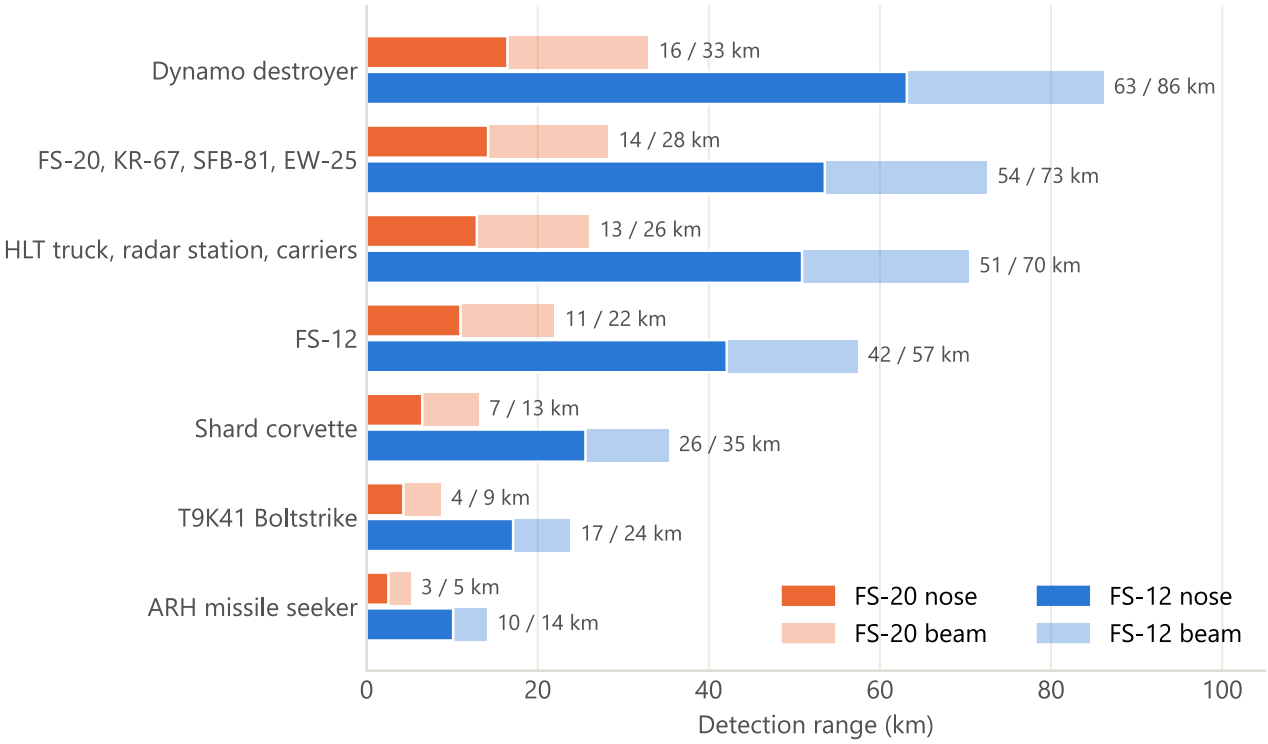
With the mod

Every radar runs the radar equation on every target: echo power against noise, clutter and jamming. A target is detected when the signal-to-noise ratio (SNR) reaches 10 dB, and no mask or notch rejects it. Ships are radar targets now, with realistic sizes.

How it works

The mod keeps vanilla ranges as its anchor. Each radar is calibrated so that a 1 m^2 target gives exactly 10 dB at its vanilla range. Then real physics takes over.

The key law: **detection range grows with the fourth root of RCS.** Sixteen times less RCS only halves the range. That is why stealth helps, and also why it does not make you invisible.



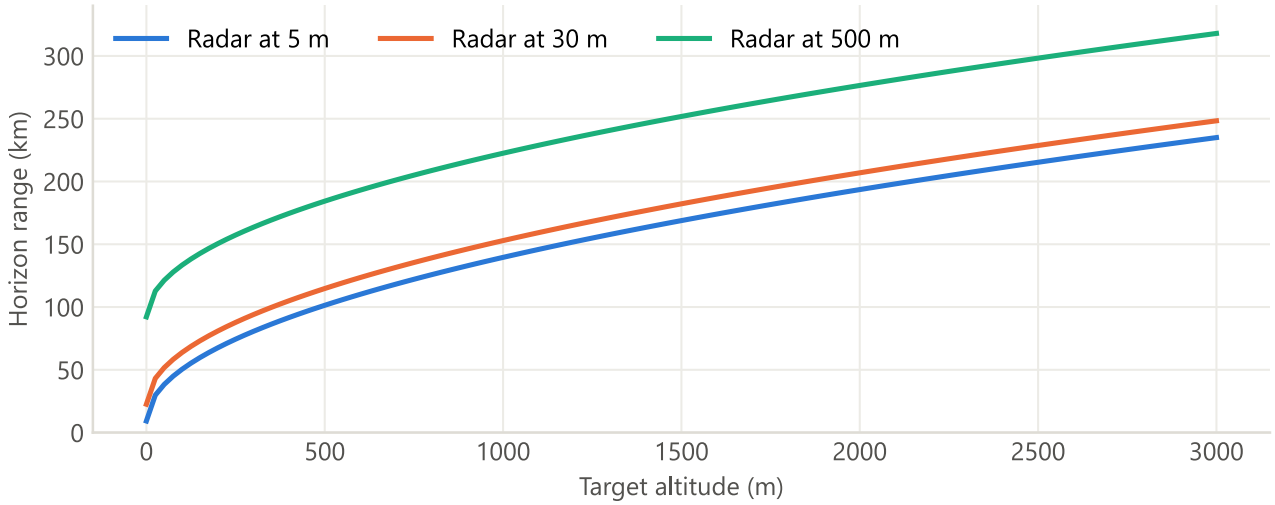
How far each radar detects a clean FS-20 and a clean FS-12, with no notch, jamming or horizon in the way. Solid bar and first number: you point your nose at the radar. Pale bar and second number: you show it your side.

External stores make you visible from farther away (chapter 5).

Radar	Range on 1 m²
Dynamo destroyer	120 km
FS-20, KR-67, SFB-81, EW-25	100 km
HLT radar truck, radar station, carriers	100 km
FS-12	80 km
Shard corvette	50 km
T9K41 Boltstrike	34 km
ARH missile seeker	20 km

Aircraft from other mods keep the range their author set.

The horizon. Radar waves bend a little in the atmosphere, so the game uses a 4/3 Earth radius, and the line of sight is traced along that bent path over real terrain and the sea. A radar near the ground cannot see a low aircraft far away, whatever its power. Terrain and ships block the path before any physics runs.



Radar horizon on a smooth Earth. Hills make the real picture better for you.

Sea reflection. Over the sea, a radar hears a low target twice: once directly, once off the water. The two echoes add or cancel. For a low radar (a ship, a SAM site, a sea-skimming missile) against a target a few metres above the waves, the sum swings between a boost close in and a deep fade toward the horizon. A ship radar on a 20 m mast picks up a 3 m sea skimmer at about 17 km instead of its horizon at about 26 km; an active seeker skimming at 5 m loses it past about 8 km. A high radar looking down at the same target does not care: from that angle the sea is rough and scatters the reflection. Ships themselves are not affected. Over land the reflection is much weaker.

Radar states. A detection has a state, which your RWR reports:

State	Meaning
Search	The radar detects you
Track	SNR is 3 dB or more above the threshold
Hard lock	You are the radar's locked target
Missile guidance	The radar guides a missile at you

What you see

The SNR bar under the max-G readout shows how close the strongest enemy radar is to detecting you (chapter 26). The ID line shows what the enemy has built from it.

How to use it

- Altitude is exposure. Every 100 m less puts you behind the horizon of a low radar sooner. A hill beats any jammer.
- Over the sea, a few metres is better than fifty. Fly very low and the sea reflection fades you out for low radars near their horizon.
- Short-range SAM radars are not long-range sensors. A Boltstrike sees a head-on FS-12 at only 17 km, and a head-on FS-20 at 4 km.
- Your own radar tells enemy aircraft where you are (chapter 7). Use it when you need it, not always.

Common mistakes

- Assuming stealth means invisible. Close enough, or from the side, every aircraft shows.
- Staying at medium altitude "just out of range". Range depends on your aspect and stores too.

Numbers and config keys

Item	Default	Key
Detection threshold	10 dB	`MinSNR_dB`
Sea reflection on/off	on	`MultipathEnabled`

The rest of section `03. Radar Physics` holds the switches for atmosphere and clutter levels.

5. Radar cross section

Vanilla

One `radarSize` per aircraft, the same from every direction, plus the RCS of stores.

With the mod

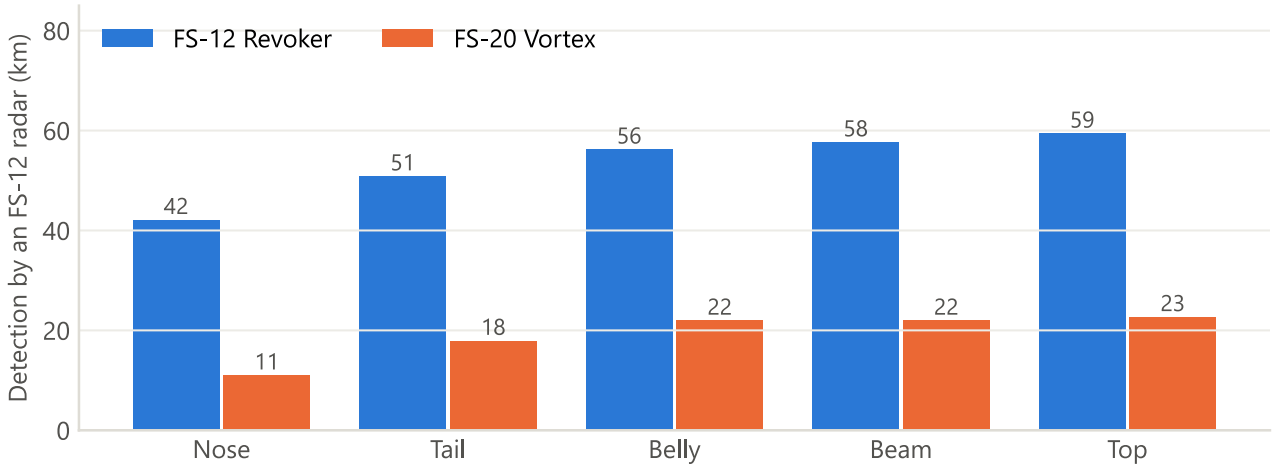
RCS depends on where the radar looks at you from. Stores, open bays and rotor blades all count.

How it works

The base is the vanilla `radarSize`. The mod multiplies it by an aspect factor, interpolated between five directions:

Airframe	Nose	Beam	Tail	Top	Belly
Conventional fighter	0.7	2.8	1.6	3.2	2.5
Stealth airframe	0.2	3.5	1.5	4.0	3.5
Large aircraft	0.75	3.5	2.0	4.5	4.0
Helicopter (plus rotor flash 0.5)	0.8	3.0	1.4	3.5	2.2

An airframe at 0.01 m^2 or less uses the full stealth profile, and one at 0.04 m^2 or more the conventional one. Between them the two blend.



The same aircraft from different directions. Turning your belly or side to a radar gives away much of what you gained.

Aircraft	`radarSize` (m^2)	Nose (m^2)	Beam (m^2)	Detection by an FS-12: nose / beam
FS-12 Revoker	0.08	0.056	0.22	42 km / 57 km
KR-67 Ifrit	0.0015	0.0003	0.0053	12 km / 24 km
FS-20 Vortex	0.001	0.0002	0.0035	11 km / 22 km

Stores. External weapons and tanks add the RCS of their mount. This hits stealth hardest: a single AAM-29 on a pylon adds 0.005 m^2, which is 25 times the frontal RCS of a clean FS-20. Internal stores behind closed doors add nothing. Open doors add a cavity (0.25 m^2 for a bay, 0.15 m^2 for an enclosed pod) plus the weapons inside, in proportion to how far the doors are open.

Helicopters. Rotor blades add a flash return and keep a helicopter out of the Doppler notch. A hovering helicopter is not hidden the way a beaming jet is (chapter 6).

Ships. The game gives all ships the same `radarSize`, so the mod uses sizes by hull class, from 25 m^2 for a stealth Shard corvette to 1 400 000 m^2 for a Hyperion carrier. A carrier is visible to the horizon.

What you see

The `RCS` line on the HUD shows your frontal RCS in m^2. It appears when the gear is locked up. Watch it as you load stores in a stealth aircraft, and as the bay doors open.

How to use it

- Fly stealth aircraft clean. Check the `RCS` line before you cross the line.

- Point your nose at the threat you worry about, and do not bank your belly or top at it.
- Open the bay only for the shot. The doors are a spike in your RCS.

Common mistakes

- Beaming a radar and expecting to stay small. The beam aspect is 3.5 to 4 times the base RCS. Beaming only works with the notch (next chapter).

Numbers and config keys

Section `03. Radar Physics`: `ExternalStoresRCSEnabled`, `HelicopterRotorModulationEnabled`.

6. Doppler notch and clutter

Vanilla

The notch is a fixed speed limit, with small effect and a helpful HUD box that gives away the missile range.

With the mod

The notch is a real filter. A radar throws away echoes that do not move relative to the ground, because that is how it removes the ground itself. If you fly across its line of sight with the ground behind you, you are thrown away with it.

How it works

Two conditions must both be true:

1. **Your radial speed relative to the ground is below 20 m/s.** Only your motion along the line from the radar counts. Fly at 90 degrees to the radar and that speed is near zero.
2. **There is clutter behind you.** The radar looks down at you, or you are below 120 m above the terrain, or you are below 300 m and the radar sees you at less than 1.5 degrees elevation.



The notch needs both geometry and background. With the sky behind you, the radar still sees you.

Jamming widens it. With ECM on, the limit grows up to 2.5 times (50 m/s at full ECM), and a jamming target in the notch is rejected at any height, even against clear sky.

Clutter outside the notch. A moving radar also sees ground clutter in a band as wide as its own speed. Below a fast fighter, a target flying away has a closing speed inside this band and is harder to see. A head-on target is outside it. A stopped radar has no such band, and an AWACS-like slow radar has a narrow one.

Who cannot notch. Ships are never rejected: sea-search modes look for them against sea clutter. Helicopters keep their rotor sidebands: only the rotor flash survives against the ground, but it does survive.

What you see

On Tier 3 aircraft, the HUD reads the situation for you. On Tier 2 you get an approximate cue. Tier 1 gets nothing.

Cue	Tier	Meaning
`[ NOTCH OK ]`	3	In the notch, and the radar rejects you
`[ ECM NOTCH ]`	3	In the notch with the jammer on
`[ NO CLUTTER ]`	3	Right geometry, but the sky is behind you. Descend
`[ BEAMING ]`	2	You fly about 90 degrees to the threat. No promise about the background

These cues need the range to the threat, from the faction estimate or the datalink. If the range is unknown, they stay off. A missile cue always replaces a notch cue.

How to use it

1. Put the radar at your 3 or 9 o'clock.
2. Get below it, or low, so the ground is behind you.
3. Hold the heading and speed along the beam. Any drift toward or away from the radar above 20 m/s takes you out.
4. In a missile fight, use the guiding radar's bearing, not the missile's (SARH), and turn hard when you enter the notch to break the missile's memory (chapter 20).

Common mistakes

- Notching above the radar with sky behind you. Nothing happens without the jammer.
- Notching the wrong radar. The notch is per radar: beaming one can point your tail at another.
- Staying in the notch with the jammer off and a second radar at a different angle.

Numbers and config keys

Item	Default	Key
Notch half-width	20 m/s	`NotchVelocityThreshold_mps`
Notch widening by ECM	1.5	`ECMNotchWidening`
Radar sidelobe clutter	-35 dB	`RadarSidelobeClutter_dB`
Seeker sidelobe clutter	-30 dB	`SeekerSidelobeClutter_dB`
Notch edge factor	2	`NotchClutterEdgeFactor`

7. RWR tiers and ESM

Vanilla

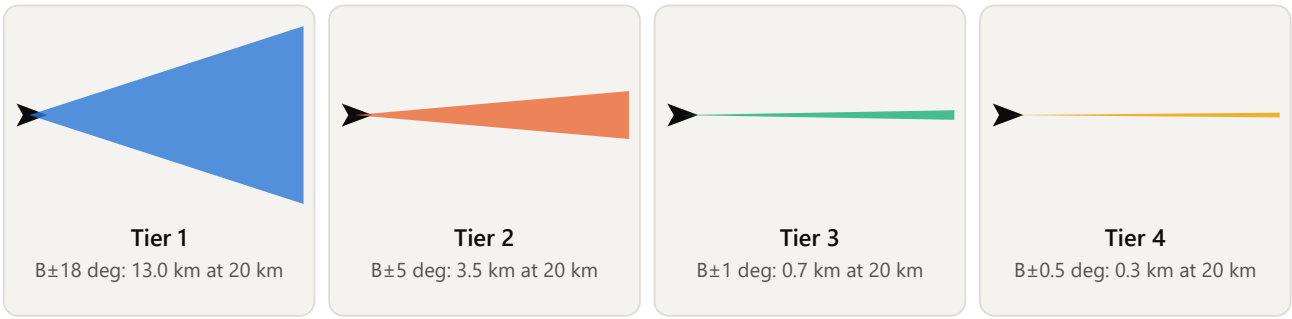
Every aircraft has the same perfect receiver.

With the mod

Your avionics generation decides what your receiver can tell you. The tier follows the rank you need to fly the aircraft, so modded aircraft work too: rank 0 and 1 are Tier 1, rank 2 is Tier 2, rank 3 and up is Tier 3. A low-rank aircraft with its own radar gets Tier 2.

How it works

	Tier 1	Tier 2	Tier 3	Tier 4 (add-on)
Who	Rank 0-1 without radar: CI-22, T/A-30, SAH-46, UH-90	Rank 2, or rank 0-1 with radar: A-19, VL-49, VT-7	Rank 3 and up: FS-12, FS-20, KR-67, EW-25, SFB-81, AB-4	Any aircraft that carries a radome, for example the EW-25. A jamming pod alone does not count
Bearing error	±18 deg	±5 deg	±1 deg	±0.5 deg, elevation 1 deg
Coverage below / above	-25 / +35 deg	-40 / +52 deg	-85 / +85 deg	Base tier
Hears an ARH seeker at	3.5 km	16 km	40 km	Base tier
ESM sensitivity	-45 dBm	-55 dBm	-65 dBm	-80 dBm
Threat classification	No	Yes	Yes	Base tier
HUD notch cue	None	`[ BEAMING ]`	Full	Base tier
Datalink	Basic	Basic	Advanced	Shares bearings
Single-ship fix	No	Yes	Yes	Yes, also on Tier 1
Optics magnification	1.0	1.5	2.0	Base tier



What one bearing means in each tier, at 20 km. Tier 1 can only tell you the quarter of the sky.

Blind cones. The elevation limits are in the aircraft's body axes. A Tier 1 aircraft is blind to radars more than 35 degrees above its nose-tail plane and 25 degrees below it. Roll or pitch and the blind cones roll with you. A ship far below your wing can vanish from the scope in a turn.

Threat states and persistence. A threat stays 3 s after its last pulse. A guidance state holds for 3 s, so one missed sweep does not clear the alarm.

ESM: hearing what cannot see you. ESM is not a separate sensor. It is your RWR working below the level of a warning. A radar needs its signal to go out and back. Your receiver needs it only one way. When a radar scans you but does not detect you (you are too far, in the notch, or jamming), your receiver can still take a bearing, if the one-way power is above your tier's sensitivity. This goes to the passive track only: no strobe, no alarm. Your RCS and your jammer do not affect it. Altitude does: low over the sea, the sea reflection also weakens what you hear from a low radar near its horizon, so those bearings can come late. A low-observable aircraft can therefore locate a radar from outside that radar's detection range.

Your own radar gives you away. Enemy aircraft carry the same receivers. When your radar scans them, they hear it, often from farther away than your radar can see them. Each bearing goes into their faction's track on you, and Q rises (chapter 11). Two enemy aircraft that share bearings can fix your position (chapter 12). AI aircraft do this too, but only human pilots make a single-ship fix. Ground units and ships have no receiver: they must find you with radar or optics.

What you see

- **Map and minimap:** strobes. Yellow search, orange track, red guidance.
- **RWR page:** symbols at the measured bearing. Closer to the centre means a stronger signal, not a shorter range.
- **EM scan ring:** a line around your aircraft on the TSD page. A bump is a received signal, a dip inward is a radar that paints you but cannot yet detect you.

Chapters 28 and 29 show all the symbols.

How to use it

- Know your tier. In Tier 1, treat every strobe as "roughly that way" and stay near allied radars.
- A strobe gives a direction only. To get a position, go to chapter 12.
- Strength is not range. A strong signal can be a weak radar close or a big radar far.

Common mistakes

- Trusting silence. A silent RWR does not mean safe: an ARH missile in midcourse makes no sound (chapter 20).
- Taking a lost SAM symbol for a dead SAM. After your ARM launch, an AI radar turns itself off so that the missile misses (chapter 22). It transmits again about 5 s after the estimated impact, at most 45 s later. Do not fly into its zone straight after the launch.
- Turning your blind cone toward a known threat in Tier 1 or Tier 2.

Numbers and config keys

Item	Default	Key
Strobe persistence	3 s	`RWRStrobePersistence_s`
Tier 1 / 2 / 3 azimuth error	18 / 5 / 1 deg	`Tier1_AzimuthJitter_deg` and siblings
Tier 4 azimuth error	0.5 deg	`Tier4_AzimuthJitter_deg`
Tier 4 sensitivity	-80 dBm	`Tier4_EsmSensitivity_dBm`

All of it is in section `06. RWR Receiver`.

8. Visual contacts and optics

Vanilla

Units spot targets with their vanilla optics, and all of the faction knows at once.

With the mod

The optics stay vanilla, but what they find becomes a precise contact in the faction track. Every unit with optics is a spotter: aircraft, ships, tanks, trucks, air defense. Aircraft see farther with better avionics: visual range is target visibility times the magnification of your tier (1.0, 1.5, 2.0).

How it works

A visual contact gives the exact position and a high Q: 1.0 at 5 km or nearer, falling to 0.90 at 15 km. A building or a unit slower than 1 m/s always gets 1.0. The contact holds for 1.5 times the spotter's check interval, so Q does not flicker between checks (ground units check every 2 to 10 s).

Jamming and the notch do nothing against eyes. Ground spotters do not see targets faster than their vanilla speed limit (200 to 500 m/s), so a fast pass can beat a truck.

Laser designator. This is not a spotter: there is no laser on ground units or ships. Only an aircraft has one, in its nose, and the vanilla rules apply:

- It lases only what its pilot has selected: 1 target on most aircraft, 2 on the CI-22 and UH-90, 3 on the A-19, 4 on the SAH-46.
- The target must be within 15 km, with a clear line of sight to the nose.
- The faction must already know the target: spotted in the last 4 s, or known to within 100 m.
- The VL-49, EW-25, AB-4 and SFB-81 have no designator.

While the laser holds, the faction track has Q 1.0 and weapons get the smallest miss. Vanilla keeps the lased state for the whole faction, so a wingman's laser weapon can use your spot too. Lose the line of sight and the laser stops at the next check, 0.25 s later.

What you see

`ID OPT` on your HUD means an enemy sees you. Your own faction's visual contacts raise Q on the marker and the map.

How to use it

- A cheap vehicle near the front is a sensor. Protect them, and kill theirs.
- If `ID OPT` shows, the notch and the jammer will not help. Get behind terrain or out of visual range.

Numbers and config keys

Item	Default	Key
Near visual Q	1.0	`VisualContactNearQuality`
Far visual Q	0.90	`VisualContactQuality`
Near range	5000 m	`VisualContactNearRange_m`
Far range	15000 m	`VisualContactFarRange_m`
Laser Q	1.0	`LaserContactQuality`

9. HMD thermal window and IRST

Vanilla

Not in the game.

With the mod

Press `B` for an infrared window on your helmet. It looks through the airframe, where your head points, and it also works as a passive sensor.

How it works

The window uses the aircraft's target camera, so it needs one, and you must be in the cockpit view. It opens at the first zoom level. `V` steps through the levels of `HMDThermalZoomLevels`: 1x, 5x, 10x by default. You can set up to three levels, each from 1x to 20x, in your own order.

Engines, rocket motors, flares, ground vehicles and ships show as white spots (not your own engines). A moving ground vehicle has intensity 0.6 and a stopped one half of that. A ship has 1.0. Buildings have no heat.

IRST. A hostile aircraft engine, vehicle or ship in the window, in line of sight and in range, gets a green bracket and sends a bearing to your faction every 0.5 s. No radar is involved and no one hears you.

- IRST range = 15 km x square root of intensity x square root of zoom.
- Bearing error = 1 degree divided by zoom.

Target	Range at 1x	At 5x	At 10x
Heat 1.0 (ship, engine at intensity 1)	15 km	34 km	47 km
Moving vehicle (0.6)	12 km	26 km	37 km
Stopped vehicle (0.3)	8 km	18 km	26 km

A 20x level reaches about 67 km on heat 1.0.

Bracket bearings work like RWR bearings, also for Tier 1 aircraft: fly a turn and they become a fix (chapter 12). This is how an aircraft without a radar can find things.

How much Q a bracket gives. Little by itself. A bearing fills Q at the slow RWR rate (chapter 11), and it does not add to the RWR rate of your own aircraft on the same target. From zero, with a Tier 2 aircraft:

Range	Q per second	Time to Q 0.5
5 km or less	0.020	35 s
15 km	0.014	49 s
30 km	0.010	70 s
47 km	0.006	110 s

Multiply the rate by 0.75 on Tier 1, 1.3 on Tier 3 and 1.6 on Tier 4. The real gain comes from the fix: Q jumps at once to match the fix error, to about 0.37 at 1 km of error, 0.66 at 300 m and 0.80 at 100 m.

What you see

A grey image with white spots, green brackets on IRST sources, and the zoom (for example `5x`) in green when it is above 1x.

How to use it

- Hold a bracket while you fly across the bearing. Zoom in for range and accuracy.
- Use it for flares and missile motors as a visual check at night.
- In a Tier 1 aircraft, this is your only single-ship fix.

Common mistakes

- Looking for a bracket on a building. Buildings have no heat.
- Expecting range from a bracket. It gives a bearing only.

Numbers and config keys

Item	Default	Key
Window key	B	`HMDThermalWindowKey`
Zoom key	V	`HMDThermalZoomKey`
Zoom levels	1, 5, 10 (up to 3, each 1-20)	`HMDThermalZoomLevels`
IRST base range	15000 m	`HMDThermalIrstRange_m`
Zoom exponent	0.5	`HMDThermalIrstZoomExponent`
Bearing error	1.0 deg	`HMDThermalIrstBearingError_deg`
Vehicle heat	0.6	`HMDThermalGroundHeat`
Ship heat	1.0	`HMDThermalShipHeat`

Part III. Knowledge

10. Tracks and memory

Vanilla

The faction database stores true positions.

With the mod

Each faction keeps one **track** for each enemy threat. A track holds a position, an error, a quality Q and a contact state. Tracks run in parallel to vanilla knowledge. HUD markers stay vanilla in how they move and lock, but tracks decide their colour, the map icons, AI launches and missile datums.

How it works

Not everything gets a track. Aircraft, ships, air defense vehicles, radars and strategic buildings do. Trucks, tanks, APCs and artillery stay on vanilla rules.

Contact	Source	What the faction gets
Precise	Radar, visual or laser	Live position, no error
Passive fix	Bearings that cross	A position with an error (chapter 12)
False fix	A track radar fooled by a decoy	The decoy's position (chapter 17)
Bearing only	One RWR bearing	A direction, no position

A precise contact stays "recent" for 4 s. A track stays in contact for 6 s after its last detection.

Memory. When contact ends, the track goes into memory. The position freezes at the last known point, Q holds, and the expected error grows with time. After the time cap, the track is forgotten:

Target	Forgotten after
Aircraft	120 s
Ground and naval units	600 s
Strategic building	600 s, or never once it had a located track
Radar building with a located track	Never, while it exists

The real movement of the target has no effect on this. A ghost does not vanish because its target moved: that would be free information. It stays until time runs out, or a new contact moves it.

What you see

- HUD markers appear when your faction has a recent contact, a located track, or Q of 0.35 or more.
- Map icons go dim when the position is old. A located strategic building stays as a `?` ghost.
- The TSD page shows dim symbols and an uncertainty ring.

How to use it

- An old track is still a target, but the target has moved. A weapon on it flies to where the enemy was.
- Against you: after a break in contact, change heading and altitude. The enemy's datum stays behind.

Common mistakes

- Reading a bright symbol as "it is there now". Check whether the contact is live.

Numbers and config keys

Item	Default	Key
Aircraft memory	120 s	`AircraftMemory_s`
Surface memory	600 s	`GroundTargetMemory_s`
Strategic building memory	600 s	`StrategicBuildingMemory_s`
Contact hold	6 s	`ContactHold_s`
HMD marker threshold	0.35	`HMDMarkerQualityThreshold`
Located threshold	0.55	`LocatedQualityThreshold`

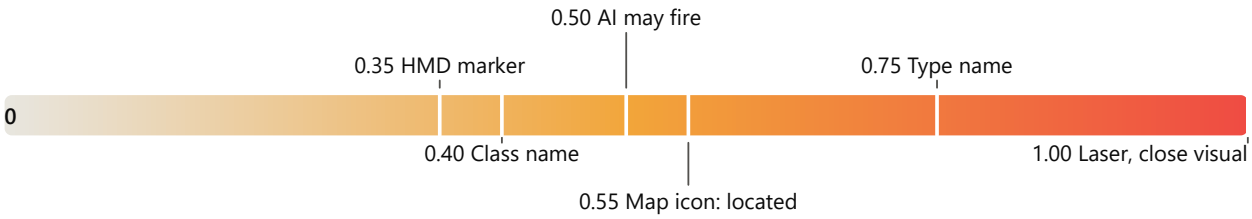
11. Track quality Q, CEP and identification

Vanilla

Not in the game.

With the mod

Q runs from 0 (unknown) to 1 (fire control). Q is how long and how well your faction has watched a target. It decides what your weapons can do and what your displays may say.

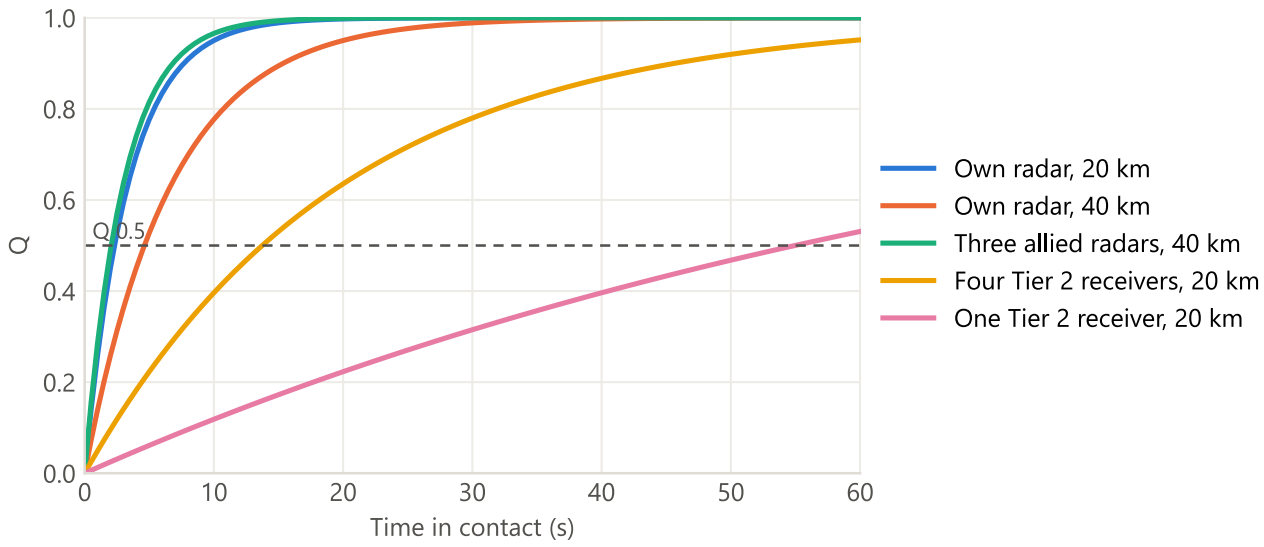


One number drives everything. Each threshold unlocks something.

How it works

Q fills. Each sensor that sees the target adds a fill rate, and the rates of the whole faction add up. Radars are fast, receivers are slow, and close is faster than far:

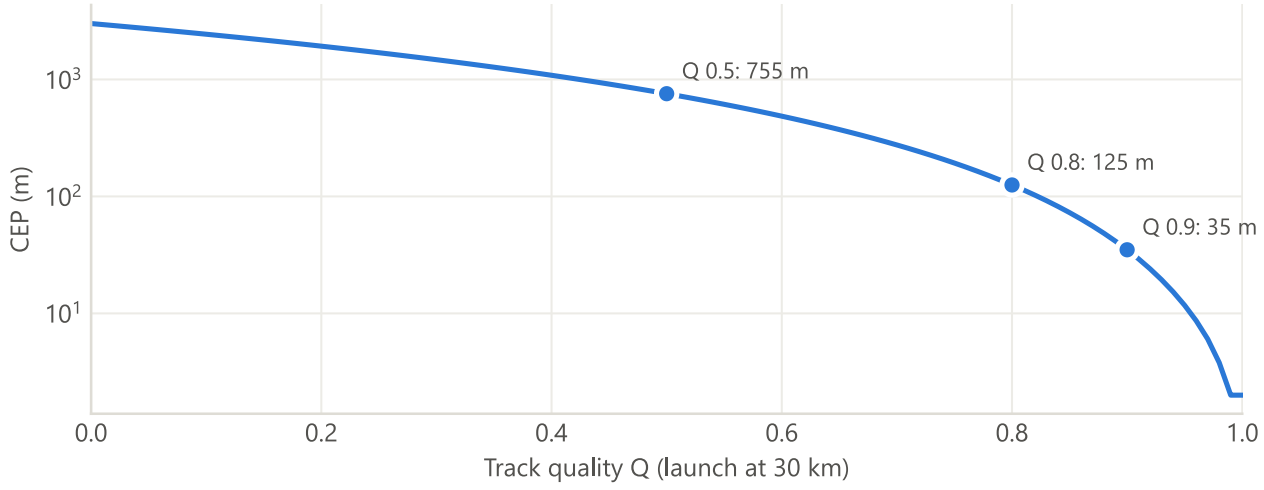
Sensor	Rate
Radar up to 20 km	0.30 per second
Radar beyond 20 km	Falls with range: half at 40 km, a fifth at 100 km
Surface radar station	x 1.5
AI or allied radar	x 0.75
RWR pulses and IRST bearings	0.02 per second x range weight x tier (0.75, 1, 1.3, 1.6)



Time to build a track. Sensors add up, so a team fills Q much faster than one aircraft.

Visual contacts, lasers and fixes do not fill slowly: they set a minimum Q at once. Q holds in memory (chapter 10). It only ends when the track is forgotten.

CEP. When a weapon launches, the mod draws one random offset from a circle. The CEP is the radius that holds half of the draws. It depends on Q and launch distance:



Weapon miss against Q. A fix at Q 0.5 is a near miss of 750 m. Q 0.9 is a hit for most warheads.

Q	CEP at 10 km	at 30 km	at 60 km
0.2	1921 m	1925 m	1935 m
0.5	751 m	755 m	765 m
0.8	121 m	125 m	134 m
0.9	31 m	35 m	42 m
0.95	8 m	12 m	17 m
1.0	2 m	2 m	2 m

The miss is horizontal. The draw is Gaussian and capped at 3 CEP. For a datalink weapon the miss shrinks in flight as the track improves. A lased target always gets the 2 m minimum.

Identification. What you may read about a target comes from Q alone. One radar paint does not reveal a type.

Tier	Q	HMD and map label	Data
Low	Below 0.40	`AIR?`, `GRND?`, `SHIP?`	Range to 10 km
Mid	0.40 to 0.75	Class: `FTR`, `ATK`, `BMR`, `EW`, `HELO`, `SAM`, `AAA`, `RDR`, `SHIP`	Range to 2 km, heading to 45 deg, altitude to 1000 m, speed to 100 km/h
High	0.75 or more	Type, for example `FS-20`	Vanilla values

Labels also show a coarse grade: `Q>0`, `Q>25`, `Q>50`, `Q>75`.

What you see

Marker tint	Meaning
Pale white	Low Q. Most weapons will miss
Amber (about 0.5)	Good for a weapon with its own seeker
Red (high)	Good for all weapons

A selected marker keeps the vanilla colour. The map selection box and the TGT page rows also follow Q and the identification tier (chapter 28).

How to use it

- Before a long shot, wait for amber at least. Keep your radar on the target while Q climbs.
- Bring friends: three allied radars fill Q in less than half the time.
- A weapon with a seeker only needs to bring the target into its basket. A weapon without one needs real Q.

Common mistakes

- Firing on a pale white marker because the range is good.
- Thinking a long bearing track is useless. RWR pulses have no Q ceiling, so patience pays.

Numbers and config keys

Item	Default	Key
Radar rate	0.30 /s	`RadarQualityRate_per_s`
Radar reference range	20000 m	`RadarQualityRefRange_m`
RWR rate	0.02 /s	`RwrStrobeQualityRate_per_s`
Mid identification	0.40	`IdentMidQuality`
High identification	0.75	`IdentHighQuality`
CEP at Q 0	3000 m	`CepAtZeroQuality_m`
Minimum CEP	2 m	`CepMin_m`

12. Passive fixes: finding a radar without radar

Vanilla

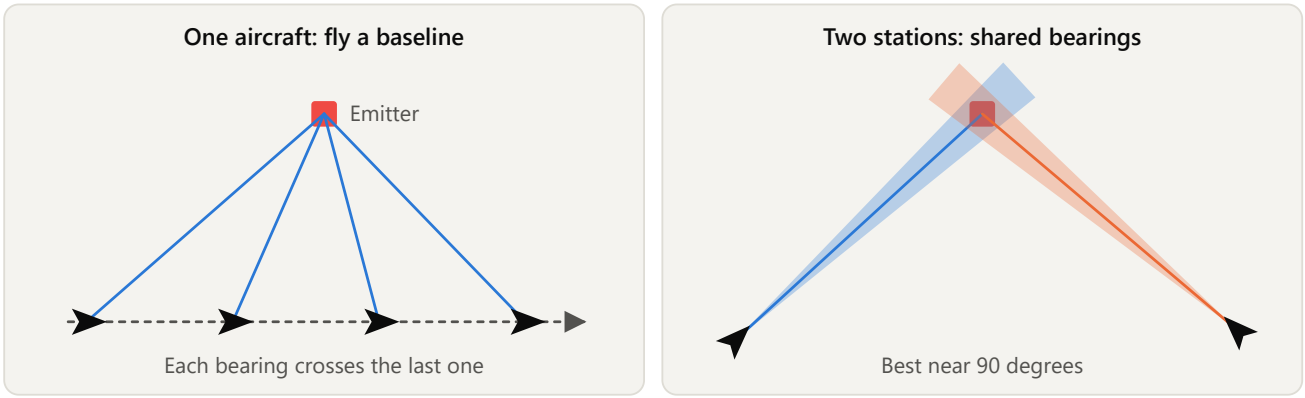
Your RWR tells the faction where the emitter is.

With the mod

A bearing is a line, not a point. To get a point, you need lines that cross. Until then the radar is not on your map and you cannot select it.

How it works

There are three ways to a fix:



Two ways to cross bearings: fly a baseline yourself, or use a second station.

One aircraft (single-ship). Fly across the bearing. Your receiver records a bearing every 0.5 s for up to 180 s, and a best fit finds the point. It starts after 3 s and 1 degree of bearing change. The error shrinks as the baseline grows. This works against stationary emitters. For a moving one, only the short window in which it barely moved counts. A Tier 1 RWR makes no single-ship fix.

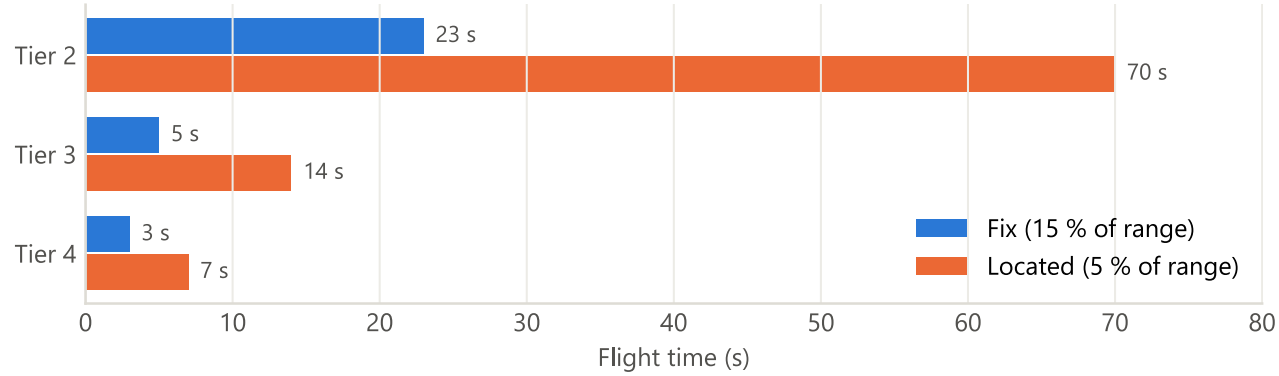
Two stations. Tier 3 and Tier 4 receivers share bearings. Two aircraft that hear the same emitter from different directions can fix it at once. Crossing near 90 degrees is best.

Tier 4 extras. A Tier 4 receiver has two more tools:

- *Elevation fix (surface emitter):* it measures elevation too, so the line of sight hits the ground at one point. It works in one pass, and it works best from high up. Below 0.5 degrees of depression it gives nothing.
- *Motion analysis (aircraft emitter):* a fit of position and velocity over 60 s. It needs a dogleg in your flight path, at least 3 % of the range off a straight line. The result is coarse, at 20 km about 7 to 9 km of error: enough for a cue, not for a weapon.

Fix and located are two different stages:

Stage	Fix error	What you get
Fix accepted	15 % of range or less	A position with an error. Q rises to match. `FIX` on the NAV page, a circle on TSD
Located	5 % of range or less	Map icon, target selectable, HMD marker



Flight time at 250 m/s to an emitter 20 km away, flying across the bearing.

The same emitter can also be located by Q 0.55 from any source, or by a radar or visual contact.

Jammers are emitters too: each jam tick gives a bearing to the jammer, with your tier's error and blind cones.

What you see

On the map, the zone circle shrinks and then turns into a narrow cone to an icon. On TSD, a dim symbol with a ring appears. As the fix gets better, the ring gets smaller.

How to use it

1. Get a strobe.
2. Turn so the emitter is abeam.
3. Hold the heading. Do not fly at the emitter: a bearing that does not change teaches nothing.
4. Watch the ring. When the icon appears, the target is yours.
5. If the radar stops transmitting, your bearings stop too. The fix stays in memory.

Common mistakes

- Flying straight at the emitter. Zero baseline, zero fix.
- Turning back and forth. A long straight leg across the bearing beats a weave.

- Expecting a fix on a fast jet. Motion ruins a single-ship fit.

Numbers and config keys

Item	Default	Key
Fix acceptance	0.15 of range	`BearingFixMaxErrorFraction`
Located fix	0.05 of range	`LocatedFixMaxErrorFraction`
Baseline memory	180 s	`SingleShipBaselineMaxAge_s`
Bearing age for two stations	6 s	`BearingObservationMaxAge_s`
Tier 4 motion window	60 s	`Tier4TmaWindow_s`

13. Datalink

Vanilla

All units share the truth.

With the mod

The datalink shares only what someone actually found. It runs a cycle every 0.35 s, and the rules are the same for AI and players.

How it works

What travels	From	To
Radar and visual contacts	Any faction unit	Faction tracks
Threat relays	Allied radars and spotters	Your RWR (up to 3.5 s old before they expire)
Hostile missile sightings	Radars of the faction that really detect the missile	Your missile warning, up to 35 km from you
Raw bearings	Tier 3 and Tier 4 receivers	Two-station fixes

Datalink	Who	Relay rate
Basic	Tier 1 and Tier 2 aircraft	Each threat at most every 2 s
Advanced	Tier 3 aircraft	Every cycle

Nodes are aircraft, ground vehicles with working sensors, and anything with a radar. A towed launcher with no sensor is not a node.

What you see

The datalink page on the small screen (`Right Shift` key) lists your nodes, nearest first (chapter 30). The map shows allied finds as icons. With `EnableDatalinkStrobes` the map and TacScreen also show cyan strobes.

How to use it

- Stay inside allied radar cover when you can. It is your best missile warning, especially in Tier 1.
- Basic datalink positions can be 2 s old. Contacts on the TacScreen are moved forward with their last velocity.
- A silent ARH missile is only visible if a friendly radar really sees it. No cover means no warning.

Numbers and config keys

Item	Default	Key
Cycle	0.35 s	`DatalinkScanInterval_s`
Basic relay interval	2 s	`BasicDatalinkRelayInterval_s`
Datalink on or off	true	`EnableAlliedDatalink`

Part IV. Electronic warfare

14. The self-protection jammer

Vanilla

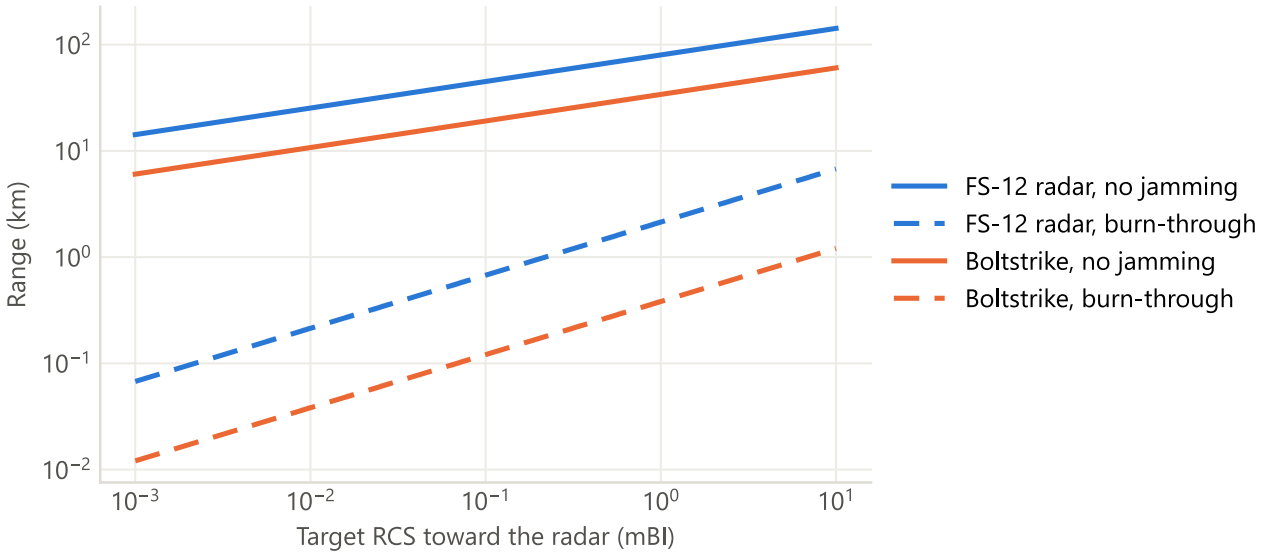
The jammer subtracts a value from the radar signal, and the engines keep it powered. It never runs dry.

With the mod

Your jammer is a noise transmitter. It adds real power to the enemy receiver. It beats a radar at long range, loses at short range, can attract a missile, and it empties your power supply.

How it works

Noise against echo. The echo travels out and back, so it falls with the fourth power of range. Jamming travels one way, so it falls with the square of range. Far from the radar, the noise wins and the radar sees nothing. Close in, the echo wins. That point is the **burn-through range**.



Burn-through range at ECM intensity 1, against unjammed detection. Jamming shrinks detection dramatically, and it helps small targets most.

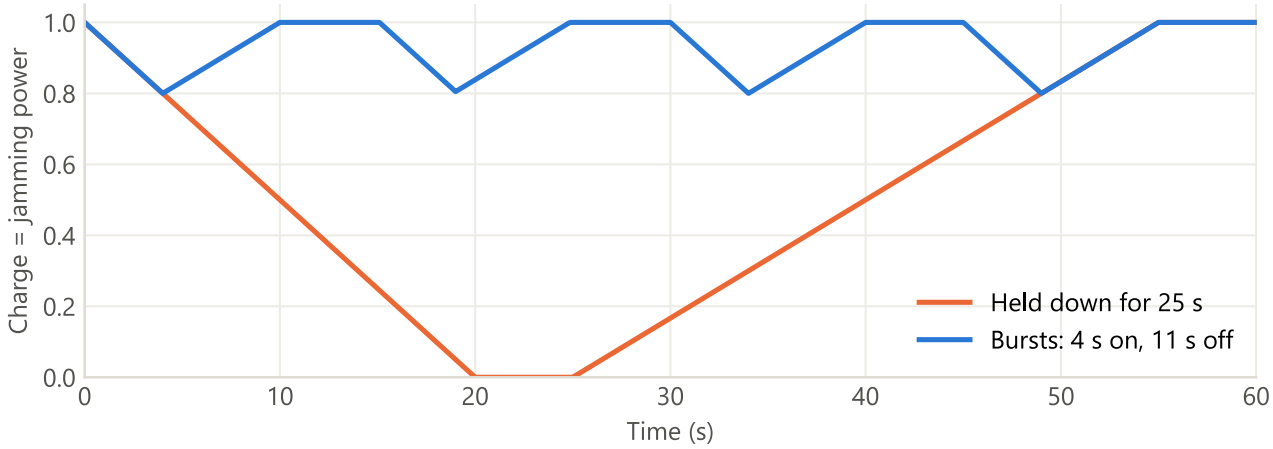
With a jammer on, detection range grows with the square root of RCS, not the fourth root. That is why a low-RCS jammer is so hard to burn through.

Home-on-jam (HOJ). A jammer is a transmitter, and a radar missile can home on it. HOJ needs all of these:

- ECM intensity of 0.15 or more.
- You are not in the notch.
- The range is more than 150 m.

The notch and the jammer work together. Inside the notch there is no HOJ, the notch is wider, you are rejected even against the sky, and the missile's memory coast is much shorter. Outside the notch, the same jammer can guide the missile to you.

Power. The jammer drains the same power supply as the vanilla battery indicator. A full supply lasts 12 s of pure drain, and about 20 s while the engines refill it at the same time. Jamming power is proportional to the charge: at 50 % you transmit at half power. The engines refill the jammer's deficit from empty to full in 30 s, and other users of the supply keep their normal power. An aircraft with a jamming pod or a laser keeps the vanilla refill speed.



Charge over a fight, with the engines refilling. Holding the button fades to nothing, short bursts stay strong.

What you see

Cue	Meaning	What to do
`[ ECM ON ]`	Jammer on, no other cue	Nothing
`[ ECM NOTCH ]`	Tier 3: jamming inside the notch	Keep it up
`[ HOJ RISK ]`	Missile guiding, jammer on, not in the notch	Notch now or stop jamming
`[ BURN-THROUGH ]`	A guiding radar is inside burn-through range	Jammer off, notch, terrain

The two red cues need the range to the threat. If it is unknown they stay off, so a quiet HUD is not a safe HUD. The countermeasure indicator shows the battery percentage.

When someone jams you, the HMD shows a band of flickering dashes in a sector around the jammer: 50 degrees wide on Tier 1, 25 on Tier 2, 15 on Tier 3. The jammer is somewhere in it, and the centre does not show its azimuth. The EM scan ring shows a wide bump.

How to use it

1. Use short bursts while the battery is high.
2. Jam at long range to deny search radars and slow down enemy tracks.
3. With a missile in the air, enter the notch first, then jam.
4. Stop when you see `[ BURN-THROUGH ]`. The noise only marks you now.

Common mistakes

- Holding the button through the whole fight. By the time you need it, you have no power.
- Flying straight at a missile with the jammer on.
- Forgetting that a jam signal gives the enemy a bearing to you.

Numbers and config keys

Item	Default	Key
Jammer power per unit of ECM	100 W	`EcmPowerPerIntensity_W`
Time to empty a full supply	12 s	`EcmJamDuration_s`
Refill from empty	30 s	`EcmRechargeTime_s`
Home-on-jam for all radar missiles	true	`HOJEnabled`
Burn-through divisor	1.0	`JammerBurnThroughRatio`

15. The stand-off jamming pod

Vanilla

The pod jams only the selected target, with a charge that builds up.

With the mod

The pod is a beam. It adds noise to every hostile radar and radar seeker inside its cone, right now, from the radar equation. Nothing accumulates. Release the trigger, and it stops 0.3 s later.

How it works

- The beam is 10 degrees wide (the EW-25 Medusa pod), aimed at your target. It is strong on the axis, down 10 dB at the edge and zero outside.
- The beam ends shortly past the target: it fades to zero at 1.25 times the target range. A radar far behind the target is not touched.
- A radar suffers most in the direction of the jammer. If the pod is in the radar's main lobe, its power is huge. If the radar looks at an escort on another bearing, jamming enters only through sidelobes.
- Terrain blocks the beam. Two pods on one target add up. The pod reaches 80 km.
- You can aim at anything the faction tracks, including a radar you found only with your RWR.

Case (Boltstrike, 100 W, 10 deg)	Result
Medusa in the main lobe at 18 km	The Boltstrike does not detect the Medusa. A RAM-45 fired at it loses the target
Escort of 3 m^2 in the cone, Medusa 50 km away on another bearing	Burn-through about 15 to 21 km, against 45 to 66 km unjammed
Escort outside the cone	Not protected

A jammed SARH illuminator has no track, so its missile loses illumination. The AI will not fire SARH through a jammed illuminator at all.

What you see

A white cone on the map and minimap for each pod of your faction. On the TSD page your own beams are amber dashes with an amber diamond at the pod target.

How to use it

- Put the enemy radar on the axis, and keep your strikers inside the cone, close to the axis.
- Escorts should fly toward the jammer's bearing from the radar's point of view.
- Hold the trigger. A pod that pauses gives the radar back at once.

Numbers and config keys

Item	Default	Key
Cone	10 deg	`JammingPodConeAngle_deg`
Pod power	100 W	`JammingPodPower_W`

16. Chaff

Vanilla

The code exists, but no aircraft carries chaff.

With the mod

Each aircraft with flares has a chaff station. A cartridge blooms into a cloud that competes with you inside the seeker's cell. If the cloud wins, the missile flies to the cloud.

How it works

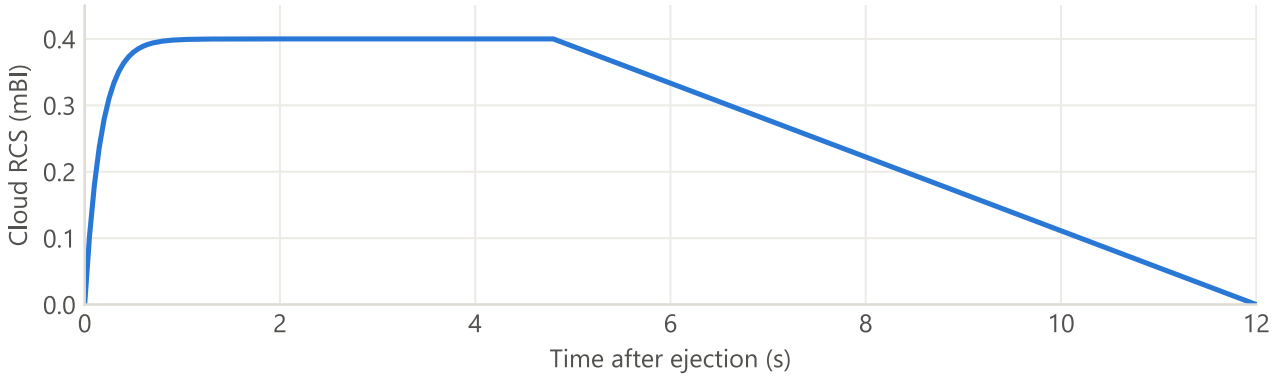
Chaff works against ARH and SARH **seekers** only. It does not affect search or track radars.

1. **The cell.** A cloud matters only if it is inside the seeker beam (8 degrees around you) and within 150 m of your range.
2. **Doppler.** Chaff stops in the air. A seeker that sees it slowing down too hard, or moving more than 25 m/s differently from you, rejects it by 30 dB. Fresh chaff decelerates far harder than any aircraft, so a new cartridge is rejected until it has slowed down.
3. **The share.** Compare what is left of the cloud against your aspect RCS. If the chaff share is above a threshold (drawn at random for each missile, 0.5 to 0.95), the seeker is captured.
4. **After the capture.** The seeker flies to the cloud until it fades, the missile passes it, or you sit in the cell with a bigger echo. It then looks straight ahead and finds you only inside its gimbal limit. So: **dispense, then turn away.**

The gimbal limit holds at all times, not only after chaff. A radar or IR missile that has flown past you, or that you have put far off its nose while it was jammed or notched, cannot see you again: it will not turn back when you stop jamming. An IR seeker loses a target or a flare more than 60 degrees off its nose (chapter 23).

A seeker in home-on-jam ignores chaff.

Case (seeker at 6 km, one cartridge)	Target RCS to seeker	Top chaff share	Capture chance
Conventional fighter, nose	0.056 m²	0.01	0
Conventional fighter, beam	0.22 m²	0.65	About 30 %
Conventional fighter, beam, two cartridges	0.22 m²	0.78	About 60 %
Stealth fighter, nose	0.0002 m²	0.66	About 35 %
Stealth fighter, beam	0.0035 m²	0.99	Almost 100 %
Stealth fighter, turning away (135 deg)	0.0015 m²	0.21	0



Chaff lives about 12 seconds, at full strength for less than five.

What you see

Silver glints and a faint haze behind you. The cockpit indicator shows `CHAFF` and the count, and the HUD shows `Radar Chaff`.

How to use it

1. Beam the missile, so the cloud and you have the same Doppler.
2. Release two cartridges, then turn away from the cloud.
3. In a stealth aircraft, chaff works even from the nose. In a conventional one, it does not.

Common mistakes

- Dispensing head-on in a conventional fighter. The seeker rejects it.
- Dispensing and flying straight. The seeker finds you again after the cloud.

Numbers and config keys

Item	Default	Key
Interval	0.75 s	`ChaffEjectionInterval_s`
Cartridge RCS	0.4 m^2	`ChaffCartridgeRcs_m2`
Lifetime	12 s	`ChaffLifetime_s`
Chaff on or off	true	`ChaffEnabled`

17. The DRFM decoy

Vanilla

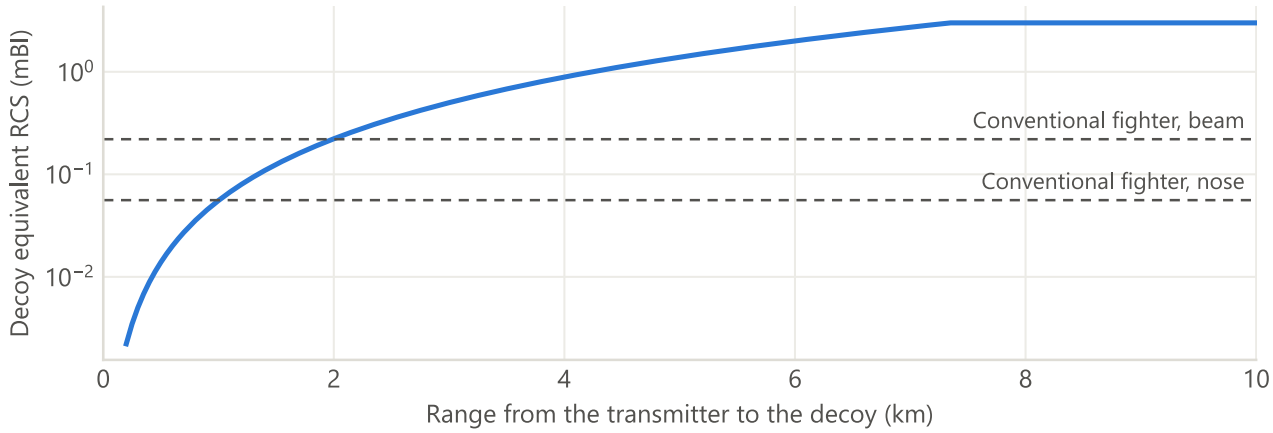
Not in the game.

With the mod

A decoy is a repeater in a flare cartridge. It copies the radar signal, with your Doppler, for 10 s. Chaff has to fight the seeker's filters. A decoy passes them. Its weakness is time: a repeater can only answer late.

How it works

- **It arms** 0.3 s after ejection, then transmits for 10 s, with a faint red flashing light.
- **It looks bigger from farther away.** Its equivalent RCS grows with the square of range from the transmitter, up to 3 m^2. Close to the missile it is weak. So release it **early**.



A decoy is strong against a far transmitter and weak against a near one.

- **Leading edge.** An ARH seeker and a track radar follow the earliest echo in their gate. A decoy behind you in range is ignored. It falls behind the aircraft, so against a missile from the front it does nothing. **It works from the side and from behind.** SARH seekers have no leading-edge tracking.
- **It beats home-on-jam.** A seeker on your jammer has no range gate, and a decoy in its beam competes with you. Your own jammer noise does not mask a decoy.

- **It can break a radar lock.** A decoy can also capture the track gate of a radar that guides a missile at you, or has one in flight. Search radars are not affected. It works when you beam that radar or fly away from it. While the gate is on the decoy:
  - The radar does not detect you, and your RWR gets no ping from it.
  - The enemy track of you shows a false fix at the decoy.
  - A SARH missile flies to the decoy, or loses the signal.
  - The AI will not fire SARH through that radar.

What you see

The indicator shows `DECOY` and the count, and while one transmits, `XMIT` and the seconds left in yellow. A third HUD line `XMIT n` shows the same whatever station you have selected.

How to use it

1. Release it with the missile several kilometres away.
2. Beam the missile or turn away.
3. Against a jam-homing missile, it is your best answer.
4. Against a guiding SARH radar, beam the radar, not only the missile.

Common mistakes

- Using it head-on.
- Using it at the last second: 1 km from the missile the decoy is only 0.056 m^2.

Numbers and config keys

Item	Default	Key
Active time	10 s	`DecoyActiveTime_s`
Interval	2 s	`DecoyEjectionInterval_s`
RCS at reference	0.5 m^2	`DecoyRcsAtReference_m2`
Reference range	3000 m	`DecoyReferenceRange_m`
Maximum RCS	3 m^2	`DecoyMaxRcs_m2`
Decoy on or off	true	`DecoyEnabled`

18. Countermeasure budget and programs

Vanilla

The flare count of the aircraft prefab.

With the mod

Each aircraft with flares carries a countermeasure budget that grows with its size and never gives fewer flares than vanilla, split your way. Two keys release a whole package.

How it works

Loadout. Three sliders sit under the fuel slider: `Flares`, `Chaff`, `DRFM`. The default keeps the vanilla flares of the aircraft and splits the rest by type. On most aircraft the vanilla flare count already fills the flare part; helicopters and attack aircraft would lean to flares if it did not. Stealth aircraft get mostly chaff, because a chaff cloud hides them well. Large aircraft that chaff cannot hide (Brawler, Tarantula) get more DRFM. AI aircraft fly with the default. A slider stops when the total reaches the budget. DRFM is limited to 15 % of the budget for everyone, players and AI: the `DRFM` row shows this limit as `max N`, and no saved file or rearm can go above it.

Countermeasure pods with their own dispensers (Aryx ECS pods, MiG-29 pods) come on top of the budget. The slider shows them as `+N` next to the count.

The budget is the larger of two values: the size of the aircraft (length times span, square root) and 1.4 times its vanilla flare count. Modded aircraft use the same rule.

Aircraft	Budget	DRFM max	Default flares / chaff / DRFM
UH-90 Ibis	90	13	64 / 20 / 6
FS-12 Revoker	90	13	64 / 20 / 6
FS-20 Vortex	101	15	72 / 22 / 7
KR-67 Ifrit	112	16	72 / 30 / 10
EW-25 Medusa	127	19	86 / 31 / 10
A-19 Brawler	168	25	120 / 23 / 25
VL-49 Tarantula	192	28	144 / 20 / 28
SFB-81 Darkreach	192	28	96 / 72 / 24

Your choice is saved per aircraft type in `BepInEx/config/CNO-EVIL.Countermeasures.txt`. AI aircraft use the defaults. Rearm refills to your choice.

Programs. A program releases types in turn: flare, chaff, decoy, flare, chaff and so on.

Program	Key	Flares	Chaff	Decoys	Interval
1	`LeftAlt` tap	1	1	1	0.3 s
2	`C`	2	2	0	0.3 s

A program uses the vanilla station path, so it works in multiplayer, and it selects your own station again at the end. A key press while a program runs does nothing. An empty type is skipped. A type never fires faster than its dispenser interval, and it needs 0.2 m of radar altitude.

How to use it

Mission	Suggested split
IR-heavy threat, low level	As many flares as the budget allows, a few chaff, DRFM 0 to a few
Radar missile threat	The vanilla flares, the rest chaff, DRFM at its `max`
Stealth strike	About half flares, most of the rest chaff, a few DRFM

DRFM never goes above the `max N` of its row (15 % of the budget). On the Revoker that is 13.

- Program 1 is your "I do not know what it is" button. Program 2 is your beam-and-turn button.
- Bind the keys to something you can reach under stress.

Numbers and config keys

Item	Default	Key
Budget at 200 m2 (length x span)	96	`CountermeasureBudget`
Budget limits	48 to 192	`CountermeasureBudgetMin` , `CountermeasureBudgetMax`
Budget floor	1.4 x vanilla flares	`VanillaFlareBudgetFactor`
DRFM share limit	15 %	`DecoyBudgetShare`
Default flare share (base)	50 %	`DefaultFlareShare`
Program 1 key	LeftAlt	`Program1Key`
Program 2 key	C	`Program2Key`
Program 1 interval	0.3 s	`Program1Interval_s`

Part V. Weapons

19. Where weapons really fly

Vanilla

Weapons fly to the true target.

With the mod

Weapons fly to a **datum**: the position your faction knows, plus the random offset from chapter 11. A seeker can then find the real target if it is inside its basket around the datum.

How it works

- At launch, each weapon gets one offset draw. The size is the CEP from Q and distance.
- If the datalink has live contact, each update rescales the same draw to the current CEP, so a better track means a smaller miss.
- In memory the datum stays frozen. No contact, no update.
- After the seeker finds the real target, the mod no longer moves the datum.
- If the track was a fix, the fix error is already in the datum.

The seeker basket. The radius around the datum in which a seeker can find you:

Seeker	Basket
ARH	`terminalRange` x tan(`maxTrackingAngle`), limited by what the seeker can detect
ARM	No limit while the radar transmits
Optical, guided bomb, guided shell	`searchRadius`
Cruise missile	`terminalSearchRadius`
Inertial shell, ballistic missile	None: only the warhead counts

Fire when the CEP is well inside the basket. With a warhead and no seeker, the CEP has to be inside the blast.

What you see

With `ShowMissileAimpointOnMap` on (a training aid, off by default), the map shows amber lines to the aim point, the offset in green or amber, and a red line once the seeker locks.

Numbers and config keys

Item	Default	Key
Dispersion on or off	true	`EnableTrackUncertaintyDispersion`

20. Active radar missiles (ARH)

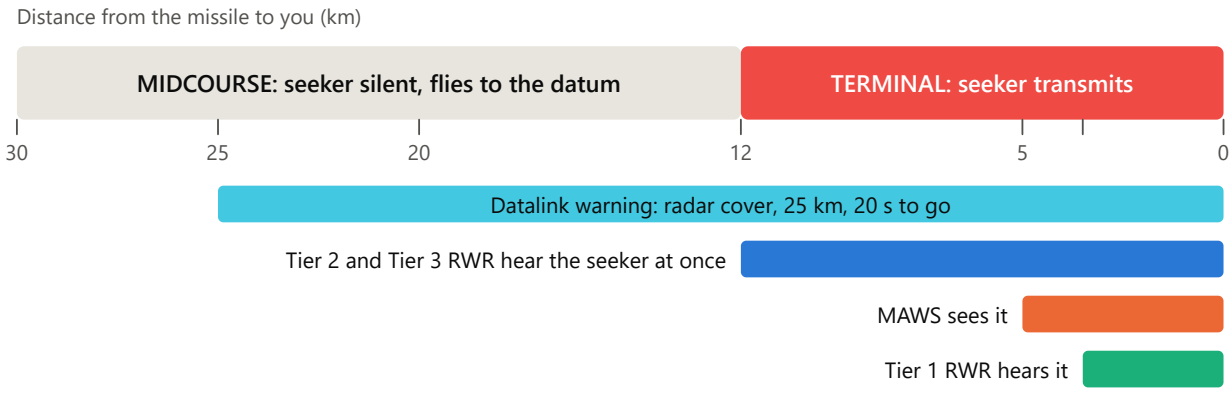
AAM-29, AAM-36, NL-98.

Vanilla

The warning comes from the missile's target data, and the seeker reads a simple signal.

With the mod

The missile is silent in midcourse and flies to the faction datum. At terminal range its seeker goes active and runs the same radar equation as any radar.



The life of an ARH missile, and what the target can hear at each stage.

How it works

- **Midcourse.** The seeker is off. The datum updates by datalink each second while your faction has a track. The offset is limited to 1800 m, because vanilla drops the target at 2000 m.
- **Pitbull.** The seeker starts at the missile's vanilla `terminalRange` (12 km from the datum). Its radar has 20 km on 1 m^2, so it may not see a small target at once, and a target in the notch at switch-on is not acquired at all.
- **Terminal.** The seeker needs SNR 10 dB. It runs the notch, clutter, jamming, chaff and decoy logic. Without a return, it flies to its own memory: the last datum moved by the known velocity.
- **Memory coast.** When the notch rejects you, the seeker keeps the lock for up to 2.5 s while you stay near the predicted path. The gate is range x tan(2 degrees), at least 30 m, and it is cut by conditions:

Condition	Usable gate
ECM above 0.1	30 %
Clutter behind you	70 %
Clear sky	100 %

There is one coast per notch entry. If you fly straight in the notch, you stay in the gate. If you turn hard, you leave it.

- **Gimbal.** The seeker sees only inside its vanilla tracking angle, in every state: search, lock, home-on-jam, memory coast, after chaff. A seeker that has lost you does not lock again inside its vanilla minimum reacquire range. A missile that overshoots you does not turn back.

What you see (as the target)

- `[ PITBULL ]` flashing red: an active seeker has you. On the RWR page, a red `M`.
- Before that, nothing from your own RWR. Only a datalink warning (chapter 24).
- Tier 2 and Tier 3 aircraft hear the seeker as soon as it transmits: their limits of 16 km and 40 km lie beyond the 12 km pitbull range. A Tier 1 aircraft hears it only at 3.5 km. A Tier 4 receiver hears like its base tier.

How to use it

As a shooter: get Q up, shoot, and keep the track alive for the datalink. A crank (turning off the nose while keeping the target in radar cover) keeps the datum fresh and slows the closure.

As a target: see chapter 32.

The HUD NEZ is kinematic only. It knows nothing about the notch, chaff, decoys or jamming, and it assumes a target altitude of 0. A target inside the NEZ can still defeat the missile.

Common mistakes

- Firing and turning cold on a poor track. Your missile keeps flying to a frozen datum.

Numbers and config keys

Item	Default	Key
Memory coast	2.5 s	`MissileMemoryCoast_s`
ARH activation override (0 = vanilla)	0	`ARHSeekerActivationRange_m`

21. Semi-active radar missiles (SARH)

RAM-45, StratoLance R9.

Vanilla

The missile follows the launcher's lock.

With the mod

The seeker receives an echo from the illuminator: radar to target, target to missile. Both paths must be clear of terrain and the horizon. The illuminator must hold its own track on you, with SNR 10 dB and out of the notch.

How it works

- Break the illuminator, and the missile has nothing. It goes on to the faction datum, then gives up after the vanilla time without track.
- The notch uses the illuminator's position. So **beam the radar, not the missile.**
- A jamming pod on the illuminator, or a decoy in its gate, also breaks it.
- A self-screening jammer can still be followed by its jam strobe under the HOJ rules.

What you see

`[ MISSILE ]` in red, a red symbol on the RWR page, an orange bump on the EM ring, and a red SNR pip.

How to use it

As the target: find the radar on the RWR page, beam it, go behind terrain. As the shooter: you must keep illuminating until impact. If the target notches you, the shot is lost.

22. Anti-radiation missiles (ARM)

ARAD-116, ARAD-45.

How it works

The seeker listens for the whole flight. A radar that scans the missile pings it, and every 4.5 s the seeker can select a pinging radar near its nose. So an ARAD launched without a lock can still find a radar, and one that lost its radar can find it again.

While the seeker hears the radar, it flies at the real radar. If the radar stops transmitting, the seeker only remembers the direction. The missile then flies to the last position plus the launch miss of your track: a low Q at launch gives a large miss. The nearer the missile was when the radar stopped, the smaller this miss: at most about 520 m at 20 km, 260 m at 10 km, 50 m at 2 km (`ARMSeekerMemoryAngleError_deg`, 1.5 degrees).

AI radars know this. When their faction sees an ARM that flies at them inside 15 km (`AIRadarShutdownRange_m`), they do not go silent at once. A SAM first tries to shoot your ARM down (`AIRadarInterceptEnabled`). While its own or linked launchers and guns can still hit the ARM in time, it keeps the radar on and fires at the ARM, as many shots as the time allows. It goes silent only when it sees that it cannot intercept in time. Then it transmits again after the estimated impact time plus 5 s, and never stays off longer than 45 s at a time. While a radar is off, it does not scan and cannot guide SARH missiles.

A SAM gives up and goes silent when:

- The ARM flies too low for its launchers (an R9 site cannot engage an ARM on the deck near it).
- You launch from too close, so the first interceptor cannot reach the ARM before it arrives. A mobile vehicle needs more lead than a fixed site: it keeps 8 s to pack up and drive away, a fixed site or ship only 3 s (`AIRadarInterceptReserveMobile_s`, `AIRadarInterceptReserveFixed_s`).
- The launchers are empty or out of envelope, or an interceptor already fired but missed too late to try again.

So an ARM against an alert SAM is no longer a free shutdown. A low, close ARM still makes it go silent at once, because it cannot be engaged. `AIRadarInterceptEnabled` = false brings back the old behaviour (silent at once).

- The ARM path must pass within 1000 m of the radar (`AIRadarShutdownMissDistance_m`). An ARM aimed at another radar does not silence this one.
- The reaction takes 0.5 to 2 s.

- After 45 s off, the radar transmits for at least 10 s, even if missiles are still near.
- Ground and ship radars use their faction's missile picture. AI aircraft use their own missile warning.
- While off, the radar gives nothing to the datalink, the faction picture or missile warning.
- Your own radar never goes silent by itself, and neither does a radar that a human player controls.

Mobile radar vehicles also move. A T9K41 Boltstrike, an HLT radar truck or any other AI vehicle with a radar that can drive packs up for 2 to 5 s after the shutdown and drives 150 to 400 m to a new site, to the side of the missile path when it can (`AIRadarRelocateEnabled`, `AIRadarRelocateMinDistance_m`, `AIRadarRelocateMaxDistance_m`). The ARM does not follow it: the seeker has no emitter, so the missile flies to the old position. A radar truck that serves launchers stays near them. The HLT radar truck lowers its mast while it drives and needs about 10 s after the stop before it can transmit; a Boltstrike can transmit on the move. Ships, buildings and fixed sites only go silent.

How to use it

- As a shooter, locate the radar first, for a good Q at launch. A shot with a poor track can miss when the radar goes silent.
- Expect the SAM to shoot back. A long, high ARM shot gives its launchers time and a clean intercept basket, so an alert site may down the missile. A low or close shot is harder for it to intercept, and it goes silent instead.
- Use the shutdown: while the SAM site is silent, it cannot see or engage you. Time your run in for that window.
- As a radar operator or fighter under attack, turning your radar off late sends the ARM to the last position with a small error. Turn it off early, while the missile is far, for a larger error, if the shooter had a poor track.
- Air defense engages a missile by its flight path, so an ARM with no target is still shot at if it flies at a defended unit.

Common mistakes

- Taking a silent SAM for a dead SAM. It comes back on after the impact.
- Firing a second ARM at the old position of a mobile radar. Wait until it transmits again and get a new fix.

Numbers and config keys

Item	Default	Key
ARM memory angle error	1.5 deg	`ARMSeekerMemoryAngleError_deg`
AI radar shutdown on or off	true	`AIRadarShutdownEnabled`
SAM engages the ARM first	true	`AIRadarInterceptEnabled`
Intercept lead, mobile vehicle	8 s	`AIRadarInterceptReserveMobile_s`
Intercept lead, fixed site or ship	3 s	`AIRadarInterceptReserveFixed_s`
Shutdown range	15000 m	`AIRadarShutdownRange_m`
ARM pass distance	1000 m	`AIRadarShutdownMissDistance_m`
Back on after impact	5 s	`AIRadarShutdownMargin_s`
Longest time off	45 s	`AIRadarShutdownMaxOff_s`
Mobile radars relocate	true	`AIRadarRelocateEnabled`
Relocation distance	150 to 400 m	`AIRadarRelocateMinDistance_m`, `AIRadarRelocateMaxDistance_m`

23. IR, optical, laser, cruise and artillery

Weapon	What the mod changes	What you do
IR missiles	Seeker vanilla, but it loses a target or flare more than 60 degrees off its nose (`IRSeekerGimbal_deg` , 0 = vanilla, no limit). The launch datum follows the track	Point and shoot. As the target, a hard turn that puts the missile past 60 degrees off its nose breaks the lock. Flares are vanilla
Optical missiles and glide bombs (AGM-48, AGM-68, AT-145)	Datum and CEP. Seeker must find the target inside `searchRadius`	Get Q up for a long shot
Cruise missiles (AGM-99, AShM-300, ALM-C450)	Datum and CEP. If no target is within `terminalSearchRadius` at terminal range, the missile flies on and detonates past the datum	Update the track while it flies
Laser weapons	Seeker vanilla. A lased target has Q 1.0 and minimum CEP	Keep the target selected, in line of sight and within 15 km
Bombs, guided shells	Launch datum plus offset	Fresh, good track at release
Guided artillery, MLRS	Datum follows the track while there is contact	Spotters make it accurate
Ballistic missile	Launch datum only. A map point keeps the aim point	Good track at launch

A self-destruct watchdog removes missiles older than 10 s that leave the battlefield (more than 120 km from the map centre, below -150 m, or above 60 km).

24. Missile warning

Vanilla

You are warned when a missile's target is you, because the game knows.

With the mod

Nobody tells you. A warning needs a sensor and a collision course.

How it works

Is it known?

Source	Condition
RWR	You hear its seeker or its illuminator (guidance state)
MAWS	Within 5 km, with line of sight
Own radar	It is a contact
Faction picture	A radar of your faction really detects it

Is it coming at you? The path must pass close to you in 90 s or less. The tolerance is your size plus a margin that grows with distance, up to 1000 m. The course must hold for 1 s. A warning ends 2 s after the last positive result. A missile you actually hear is warned at once.

The datalink gate. A path you know only from a radar also needs:

- Time to go of 20 s or less.
- Distance of 25 km or less, 5 km for an IR missile.
- A seeker kind that is switched on: ARH, SARH, IR by default; ARM, optical, laser off.
- Line of sight from the missile to you.

What you see

Sign	Meaning
Normal alarm	Your RWR or MAWS sees the missile
Alarm at lower pitch	Only the datalink sees it
Cyan `M` on the RWR page	A warned missile you do not hear yourself, at its known bearing
Red `M`	Your RWR hears the seeker
Two orange arcs on the RWR page	The nearest missile that comes at you
Warning lights	Each missile warning, and each missile guidance signal

With `EnableMapDatalinkMissiles` on, the map shows diamonds with a 10 s velocity line.

How to use it

- A low alarm is a gift: you have up to 20 s. Use it to beam and descend before the seeker wakes up.
- No radar cover means no datalink warning.

Numbers and config keys

Item	Default	Key
Course hold	1 s	`MissileWarningHold_s`
Datalink time to go	20 s	`MissileWarningDatalinkTimeToGo_s`
Datalink range	25000 m	`MissileWarningDatalinkRange_m`
IR datalink range	5000 m	`MissileWarningDatalinkIRRange_m`
Picture coast	3 s	`MissilePictureCoast_s`
Datalink alarm pitch	0.75	`DatalinkAlarmPitch`

Part VI. The AI

25. How the AI thinks

Vanilla

The AI fires when the target is known and in range, and it reacts to missiles aimed at it because the game tells it.

With the mod

The AI uses the same tracks and the same warnings as you. It has no secret knowledge. Only the vanilla effects of pilot skill (reaction time, error) stay.

How it works

Launch doctrine. For a guided weapon at a tracked threat, the AI holds fire unless:

1. Its faction has a track.
2. Q is 0.5 or more.
3. A SARH shooter's illuminator transmits and is not jammed or decoyed.
4. For a datum weapon, hit probability is 0.5 or more: the chance that the offset lands inside the seeker basket or the blast radius.

SARH, laser and IR weapons skip the last check. The rules apply to aircraft, SAM sites, ships, vehicles and guided artillery. Unguided weapons and untracked targets stay vanilla.

Air defense against missiles. It does not know a missile's target. It looks at the flight path, finds who is under it, and defends. A missile that flies at a defended unit is engaged.

Evading. AI aircraft react to their own missile warning.

- They beam a radar missile, dropping to 10 m above the terrain at full throttle.
- Below 8 s to impact, they fire the selected countermeasure.
- Station choice: decoy (once per missile, if more than 60 degrees off the nose), then jammer if in the notch and charged, else chaff, else jammer.
- After chaff or a decoy they turn away from the cloud for 6 s.
- Against IR they use flares and idle throttle.

Attacking. With its own missile in flight, the AI cranks 50 degrees off its target, then turns back if it wants another shot.

Radar warning. When an AI aircraft feels detected it descends, not below its minimum height.

Radar silence. An AI radar that sees an ARM flying at it within 15 km goes silent until about 5 s after the impact, at most 45 s (chapter 22). While silent it gives nothing to the datalink and guides no SARH missile.

Air defense layout. At each mission start the host builds a new layered air defense for every faction from the mission's own units. It happens once: units do not move to new sites during the mission.

- **Nodes.** The defense forms around active airbases and groups of factories, depots and ammo stores.
- **Posture.** Each faction draws one at random: Forward (wider ring, toward the enemy), Deep (closer ring, wide arc) or Sector (one flank).
- **Core and outposts.** Point defense (AAA, close-in guns, lasers) stands inside the base or on the runway approaches. IR SAMs stand outside the base, or 300 to 800 m from a radar SAM as its escort. Radar SAMs and radar stations stand farther out. An outpost moves up to 6 km, a radar station up to 15 km, often to a hilltop.
- **Sites.** Flat ground, no hollow, a clear view for every radar and optical sensor, clear of roads, runways and taxiways, and for launchers and stations clear of trees. A battery keeps its trucks and stays together. If no site fits, it keeps its mission-file position.
- **What never moves:** units named by an objective or outcome, units with waypoints, units on or next to buildings or ships, units of the map scene, tutorials and missions started from the editor.

The F10 switch does not move units back. Clients see the layout of the host.

How to use it

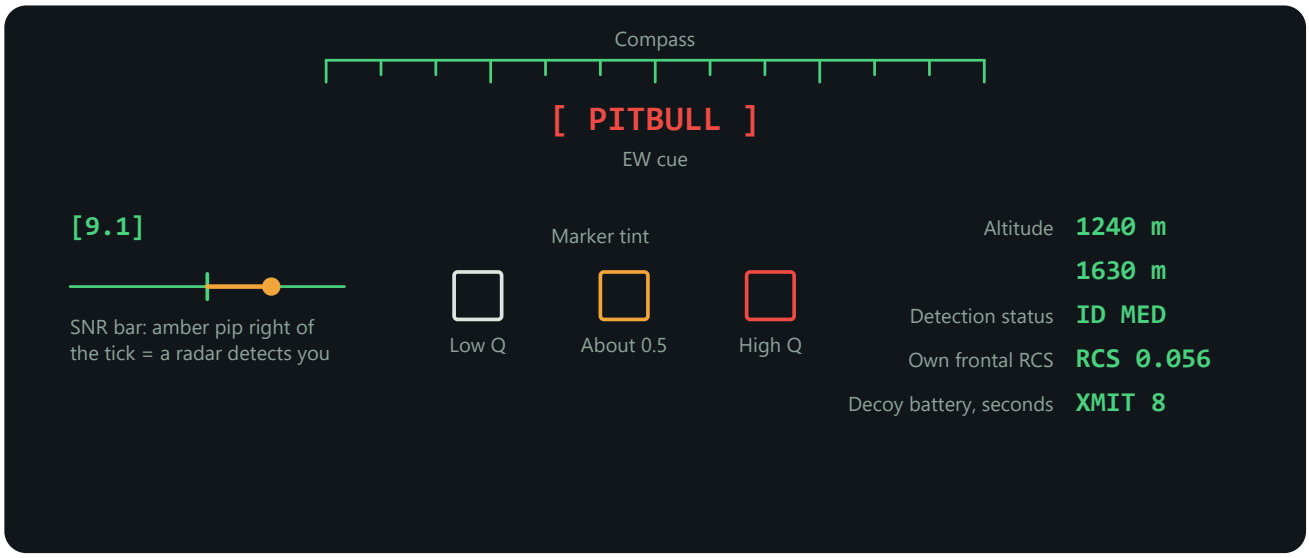
- Deny Q. An AI that cannot reach 0.5 does not fire. Notch, jam, stay at the edge of its detection.
- A decoy in a radar's track gate stops SARH shots through that radar.
- Kill sensors first. A SAM without a track from its own radar or a friend is a spectator.
- The AI's chaff is real. Your missiles can be seduced too. Attack with two.

Numbers and config keys

Item	Default	Key
Minimum Q	0.5	`AIMinLaunchQuality`
Minimum hit chance	0.5	`AIMinLaunchHitProbability`
Crank	50 deg	`AICrankAngle_deg`
Turn away	6 s	`AITurnAwayWindow_s`
AI drag manoeuvre	false	`AIDragEnabled`
Radar silence against ARMs	true, 15000 m, at most 45 s	`AIRadarShutdownEnabled`, `AIRadarShutdownRange_m`, `AIRadarShutdownMaxOff_s`
Dynamic placement	true	`DynamicPlacementEnabled`
Core radius	4000 m	`DynamicPlacementCoreRadius_m`
Outpost move	6000 m	`DynamicPlacementOutpostRadius_m`
Radar station move	15000 m	`DynamicPlacementStationRadius_m`
Gap between units	3 to 15 m	`DynamicPlacementMinGap_m`, `DynamicPlacementMaxGap_m`
Seed (0 = new layout each start)	0	`DynamicPlacementSeed`

Part VII. Cockpit

26. HUD



Where the mod's HUD items sit. Positions are schematic.

EW cue

One cue on the compass for the most dangerous threat. Top of the table wins.

Cue	Colour	Meaning	What to do
`[ BURN-THROUGH ]`	Red, flashes	A guiding radar sees through your jammer	Jammer off, notch, terrain
`[ HOJ RISK ]`	Red, flashes	A missile can home on your jammer	Notch now
`[ PITBULL ]`	Red, flashes	Active seeker on you	Notch, turn hard, descend
`[ MISSILE ]`	Red, flashes	A radar guides a missile at you	Notch that radar
`[ ECM NOTCH ]`	Green	Tier 3: in the notch, jamming	Hold
`[ NOTCH OK ]`	Green	Tier 3: in the notch, rejected	Hold
`[ NO CLUTTER ]`	Amber	Tier 3: sky behind you	Descend
`[ BEAMING ]`	Amber	Tier 2: about 90 degrees to the threat	Hold, descend
`[ LOCK ]`	Amber	A radar tracks you	Prepare
`[ ECM ON ]`	Dim green	Jammer on	None

Flashing cues alternate forms, such as `[ PITBULL ]` and `[ >> PITBULL << ]`.

Status lines

Line	Meaning	What to do
`ID NO`	No hostile track on you	
`ID MEM`	They lost you and remember a position	Change heading and altitude. Stay quiet
`ID LOW`	They know something flies here (`AIR?`)	Stop transmitting
`ID MED`	They know your class	Prepare to defend
`ID HIGH`	They know your type	Defend
`ID OPT`	Someone sees you	Terrain or distance. ECM does nothing
`RCS n`	Your frontal RCS in m^2, after the gear locks up	Check your stores
`XMIT n`	Seconds left on your active decoy	

The ID line is a deliberate exception to "you only know what your sensors know", so you can learn. It is the best track of any enemy faction.

SNR bar

Below the max-G readout. The scale is -20 dB to +20 dB around the detection threshold.

Pip	Meaning
Left of the tick	A radar paints you but cannot detect you
At or right of the tick, amber	The radar detects you
Red	The strongest radar guides a missile
Fades left	No signal

Active seekers do not show here.

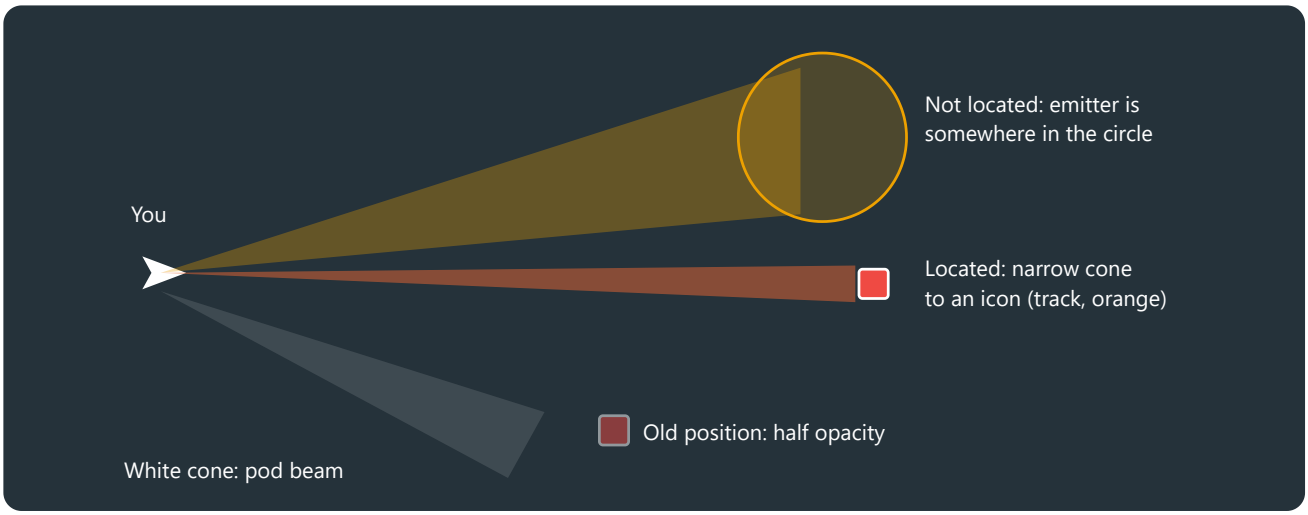
Markers

Vanilla markers, tinted by Q (chapter 11). The vanilla notch box with exact missile range is gone.

27. HMD

- **Target info** shows the identification tier: `AIR?`, `FTR`, `FS-20`, with the grade and rounded data (chapter 11).
- **Jammer cue**: 72 dashes flicker in a sector around each jammer that jams you, with no text. The width depends on your tier. The vanilla marker that showed the true jammer position is hidden. A jammer in your blind cone gets no cue.
- **Thermal window**: chapter 9.
- **MFD copies**: chapter 29, "Pages on the HMD".

28. Map and minimap



Map symbols. Colour tells threat level, shape tells how much your faction knows.

Symbol	Meaning
Yellow, orange, red strobe	Search, track, guidance
Cone ending in a circle	Not located: the emitter is at a random point in the zone, not at the centre
Narrow cone to an icon	Located
Full-opacity icon	Live or recent position
Half-opacity icon	Old position
No icon	Not located
White cone	Jamming pod beam of your faction
Cyan strobe	Datalink report (`EnableDatalinkStrobes` , off by default)
Diamond with a line	Hostile missile, 10 s ahead. Cyan, or flashing red if it flies at you (`EnableMapDatalinkMissiles` , off by default)

A strobe fades 3 s after the last signal. The tooltip and the selected-target panel follow the identification tiers. `]` and `[` zoom the minimap.

Selection box. The white box with name, speed, altitude, heading and range appears only when your faction's Q on the target reaches 0.55 (`LocatedQualityThreshold`), even if a radar sees it right now. The icon can show while the box does not. From 0.55 the box shows the data of the identification tier, and the pilot name only at High.

TGT page. Below the High tier, a target row shows the HMD label with the Q grade (for example `SAM Q>50`), `?` for the grid, and a range rounded to the tier. At High it shows the full name and the vanilla grid and range.

Filters. The type buttons (SAM, RDR and others) act on a tracked target only at High. GND and BLD act from Mid. AIR and SHP work as in vanilla. Below High, a type filter does not stop you from selecting a tracked threat.

29. PortalMFD pages

PortalMFD is the default TacScreen. It shows two pages side by side, and modded cockpits can show up to four. `;` and `'` change pages, `/` and `.` serve extra screens, and Left Shift plus the key changes the range (`PortalMFDRangeModifierKey` sets another key). The order is TSD, RADAR, RWR, EOTS, SMS, NAV, HSI, and a page shown elsewhere is skipped. F10 returns the vanilla screen.

TSD: tactical situation

Heading-up plan view: 5, 10, 20, 40 (default) or 80 km.

Symbol	Meaning
White arrow	You
Red	Hostile, located
Blue	Friendly aircraft and ships
Hat	Air contact (half ring for friendly air)
Diamond	Ship
Square	Hostile ground unit
Dim, with a ring	No live contact. The ring is the uncertainty
Green circle	Selected target
Amber dashes, amber diamond	Your pod beams and pod target
Red dashes	Radar locks of your faction, brighter while guiding
Green lines and arc	Your own radar coverage

The EM scan ring runs around your aircraft:



EM scan ring: the shape tells you what the sky is doing, without any range.

Line	Bump outward	Dip inward
Yellow (middle)	Search or track radar. A wide bump is a jammer	A radar paints you but cannot detect you. Deeper is weaker
Orange (outside)	A radar guides a SARH missile	
Red (inside)	An active seeker tracks you	

The biggest bump is always full height, so compare bumps only with each other. Small twitches are receiver noise.

RWR



RWR page symbols. Distance from the centre is signal strength, not range.

Symbol	Meaning
Green letter	Ground: `S` SAM, `A` AAA, `R` other radar. Ship: `N`. Aircraft: first letter of the class label (`F`, `A`, `B`, `E`, `H`). `U`: no classification (Tier 1)
`M`	Missile
Hat	Airborne emitter
Both half rings	It tracks you
Top half ring	Newest contact
Diamond	Strongest signal
Red	Guiding a missile (held 1 s)
Blinking lock line	A lock on you
Yellow arc, 5 s	Missile launch on that bearing
Two orange arcs	Nearest missile
Cyan `M`	Warned missile you cannot hear
`RWR OFF`	Receiver not working

RADAR

B-scope: bearing across, range up, ±60 degrees, 10 to 80 km. Orange triangles are hostile, and blue rings are friendly. The number is altitude in thousands. Dim bands lie outside your scan limits. The lock box shows bearing `B`, range `R`, altitude `A`, closure `Vc`. Short green strobes mark a jammer, within ±60 degrees of the nose with your radar on. Without a radar, the page says `OPTICAL` or `NO RADAR`.

Other pages

Page	Content
EOTS	Target camera. `CAM OFF` when off. Can open by itself when the camera comes on
SMS	Stations around your aircraft. Rounds in orange, `RLD` reloading, `SAFE`, `ARM`, `NO STORES`
NAV	Level map, heading-up, contacts with names, `FIX` on unlocated fixes, airbases in faction colours. 5 to 80 km
HSI	Compass, amber needle to the nearest friendly base, green track bug, fuel (amber below 20 %), speed, altitude

Pages on the HMD

On a small screen or at a low resolution the cockpit MFDs are hard to read. Set `EnableHMDPortalPages` to true and copies of your MFD screens float on the helmet display, in the cockpit view only. Each of four windows copies one cockpit screen (`HMDPortalPage1Slot`: Left, Right, Center, Lower, or None) and has its own position and size (`X`, `Y`, `Size`, fractions of the screen). By default the left screen sits bottom left and the right screen bottom right. The page keys change the cockpit screen and its HMD copy together. `HMDPortalPagesKey` can show and hide them (None by default: always shown while enabled), and `HMDPortalPagesOpacity` sets how solid they are. The windows have the same size and shape in every aircraft, whatever the size of its cockpit screens: the page is drawn again for the window, sharp at your screen resolution. `HMDPortalPagesAspect` (1.41) sets the width as a multiple of the height. The EOTS page is the exception: it holds the live camera image, so the window shows the cockpit screen picture, fitted inside the window.

Item	Default	Key
HMD pages on or off	false	`EnableHMDPortalPages`
Show / hide key	None	`HMDPortalPagesKey`
Opacity	0.85	`HMDPortalPagesOpacity`
Window width / height	1.41	`HMDPortalPagesAspect`
Window 1	Left screen, X 0.14, Y 0.25, size 0.35	`HMDPortalPage1Slot`, `...X`, `...Y`, `...Size`
Window 2	Right screen, X 0.86, Y 0.25, size 0.35	`HMDPortalPage2...`
Windows 3 and 4	None (X 0.14 / 0.86, Y 0.65)	`HMDPortalPage3...`, `HMDPortalPage4...`

Size is a fraction of the screen height, and Y counts from the bottom. A window shows only a screen that the cockpit has, and only with the PortalMFD preset. On cockpits with a centre or lower screen (for example the VL-49, F-16M, F-22E, F-99, Mirage 2000D), set window 3 or 4 to Center or Lower to copy it. On cockpits with no right screen (the VL-49, FS-3), window 2 stays hidden.

30. The small cockpit screen

The small flight instrument screen shows an attitude display by default: blue sky, brown ground, a ladder every 5 degrees with labels every 30, a heading strip at the top, speed at the left and radar altitude at the right.

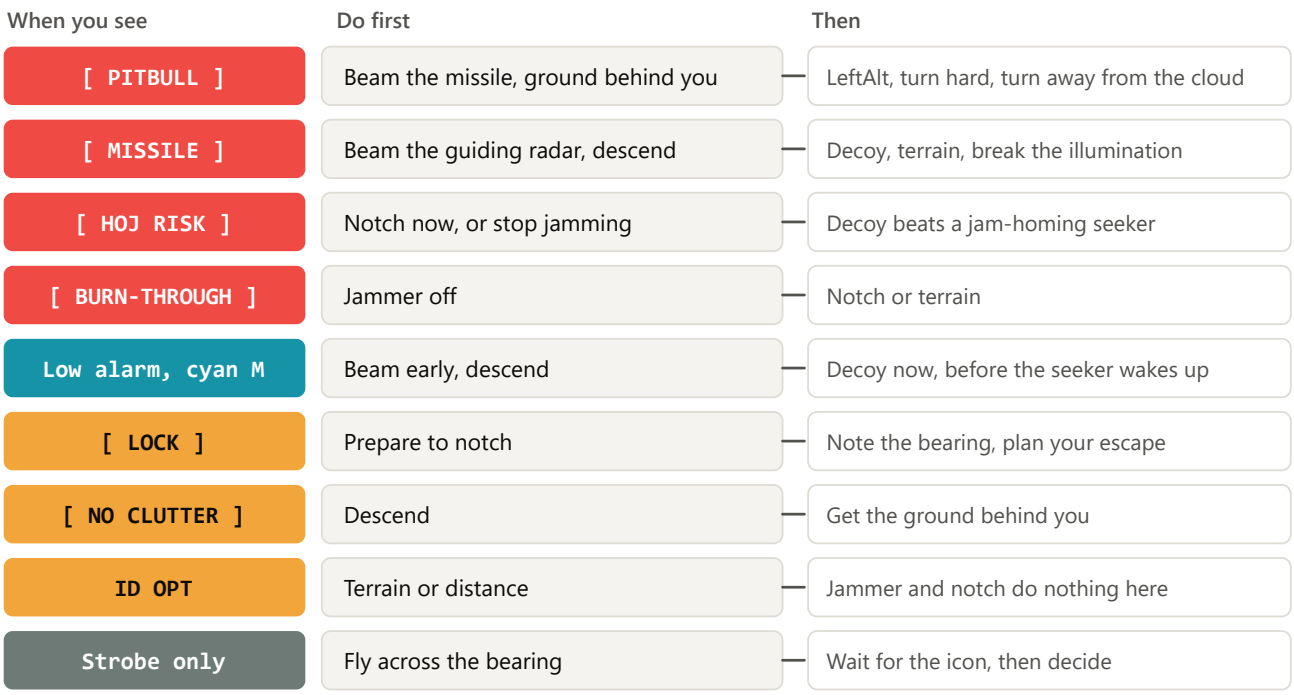
A tap of `Right Shift` changes it to the datalink page, a list of up to 16 friendly nodes, nearest first, with distance in km (nm with imperial game units). White nodes have a working radar.

Item	Meaning
`DL BAS`, `DL ADV`	Your datalink tier
`BAS+EL`, `ADV+EL`	With a Tier 4 receiver
`NODE`	All friendly nodes
`TX`	Hostile contacts your own sensors see now
`RX`	Other located contacts from the network
`+N MORE`	The list does not fit
`NO NODES`, `LINK OFF`	Nobody there, or datalink off

Some modded cockpits keep their own instruments. The Su-30 and cockpits whose flight screen shares graphics with another panel show the vanilla instruments. The MiG-29 and MiG-29SMT have no such screen. On these, `Right Shift` does nothing.

Item	Default	Key
Attitude display on or off	true	`CockpitAttitudeDisplay`
Page at start (Attitude or Datalink, saved)	Attitude	`CockpitDisplayPage`
Page key (tap)	RightShift	`CockpitDisplayPageKey`

Part VIII. Procedures



What to do when something happens. Keep this page handy.

31. Find and kill a SAM site

Do not trust your memory of the last sortie. Each time a mission starts, each side builds a new layered air defense from the same units. Guns stand at the base and on the runway approaches, IR SAMs around it, and radar SAMs farther out. The side can lean forward, stay close around the base, or favor one flank, so the picture changes from start to start. Outposts stay outposts: a lone radar SAM stays forward of the base, and a radar station can be up to 15 km from its old place with its cover. Units choose flat ground with a good view, never a hollow. An R9 battery and its trucks stay together. Units that a mission objective names stay where the mission put them, and so do units with waypoints. Tutorials and missions started from the editor do not change. The layout is set once at mission start (chapter 25). To turn this off, set `DynamicPlacementEnabled` to false. To get the same layout at every start, set `DynamicPlacementSeed` to any number other than 0.

1. **Listen.** Fly toward the area with your radar off. Note the strobe.
2. **Stay low and out of range.** Short-range SAM radars need a clear line.
3. **Fly across the bearing.** Tier 2: about 70 s for a located fix at 20 km. Tier 3: about 14 s. Tier 4 from high up: one pass.
4. **Check the map.** A zone becoming an icon means located. Check the marker colour.
5. **Pick the weapon.** ARM from outside its range, or a seeker weapon once the CEP fits the basket.

6. **Shoot and turn.** Leave the bearing, and keep your receiver on the emitter while the weapon flies. A SAM that sees your ARM coming goes silent for up to 45 s, and your ARM then flies to its memory point. A good Q at launch makes that point accurate.
7. **If it stops transmitting**, the track holds for 600 s. The site does not move during the mission, but at the next start it will be elsewhere.

32. Survive a radar missile

ARH, seeker silent (low alarm, cyan `M`):

1. Turn the missile to your beam, and descend.
2. Drop a decoy now, if it is not coming from your nose.
3. Do not jam until you are in the notch.

ARH, `[ PITBULL ]`:

1. Beam it and get the ground behind you.
2. Tap `LeftAlt`. Press `C` if you have chaff to spare.
3. Turn hard as you enter the notch: this breaks the memory coast.
4. Turn away from the cloud.
5. In the notch, jam in short bursts.

SARH, `[ MISSILE ]`:

1. Find the guiding radar: red on the RWR page, orange bump on the EM ring.
2. Beam the radar. Go behind terrain if you can. No path, no missile.
3. Release a decoy while beaming the radar to break its gate.

`[ HOJ RISK ]`: notch, or stop jamming. A decoy beats a jam-homing seeker.

33. SEAD as a pair

The model rewards two pilots who split the job.

1. **The Tier 3 or Tier 4 aircraft leads.** Both fly apart, 90 degrees across the emitter's bearing.
2. **Shared bearings** make a two-station fix in seconds.
3. **The jammer** (EW-25 with a pod) puts the radar on the beam axis, and the shooter flies inside the cone.
4. **The shooter** fires when the marker is amber or better.
5. **Spotters** on the ground raise Q for free.

34. Deep strike in a Tier 1 aircraft

You have no radar, no single-ship RWR fix, ±18 degree bearings, and a blind sky.

1. Plan for terrain. Stay below every radar horizon you can.
2. Stay in allied radar cover as long as possible: the datalink is your warning.
3. Use the thermal window: an IRST bracket gives bearings and a fix. A ground vehicle has heat.
4. Use the low alarm. At 3.5 km your own RWR will be too late.
5. Load flares and decoys. Without a notch cue, beam and descend by habit.
6. In a helicopter you cannot notch, so terrain is your only shield.

35. Notes by aircraft type

Type	Strengths	Watch out
Stealth fighter (FS-20, KR-67)	Tiny frontal RCS, Tier 3 receiver, full notch cues, chaff works even head-on	External stores wreck it. Beam and top views are 15 to 20 times the nose
Conventional fighter (FS-12)	Tier 3, decent radar	Seen at 42 km by its own type. Chaff only from the beam
EW-25 Medusa	Tier 3, Tier 4 with the radome, pod beam	Pod aims at one target, and it is a big emitter. The radome adds 0.05 m^2
Bombers (SFB-81, AB-4)	Stealth profile, Tier 3	Open bays are a beacon
Attack and trainers (CI-22, T/A-30)	Cheap, low	Tier 1: blind and late
A-19 Brawler, VL-49 Tarantula, VT-7	Large countermeasure budget, more DRFM	Tier 2: ±5 degree bearings, basic cues
Helicopters (SAH-46, UH-90)	Terrain masking, low	Cannot notch, Tier 1

Part IX. Reference

36. Multiplayer and other mods

Multiplayer

- All players need the same version. The host runs radars, tracks, missiles and AI. Clients get their faction's tracks, RWR feed and missile state.
- If the host does not have the mod, it stays off on the client, and the log says so. Version 1.1.5 uses mod protocol 12: a client with another version runs without the mod, and the host log says so.
- The host places the air defense at mission start. Clients see its layout, and its radars going silent against ARMs.
- The host decides all values that change mechanics or what a player knows. A client can change only looks and keys.
- Chaff and decoys work only when all players run the mod.
- F8 to F11 work only on the host. Clients see `DEBUG KEYS: HOST ONLY`.

Other mods

Mod	Status
Aircraft mods	Physics needs no list: radar and RCS come from the aircraft's data, the RWR tier from the rank needed to fly it
Aryx Weapons Pack	ECS flare pods add flares on top of the budget. AEC-124 jams surface targets only
Balance and Variety Rework	Works. The countermeasure budget wins over its flare count. Its flare rejection applies
MiG-29 Fulcrum, MiG-29SMT	Works. The budget sets the jet's own flares and chaff; its countermeasure pods come on top. Its own chaff model is off while CNO-EVIL is on: chaff follows the CNO-EVIL physics. Its chaff programme still applies to its chaff key, not to the CNO-EVIL program keys
Mirage 2000D	Works. Its Spirale dispenser is the chaff station and gets the chaff share of the budget. Its RWR page and radar scope show CNO-EVIL data while the mod is on. Its SEMI and AUTO countermeasure modes and its voice callouts still react to the vanilla missile warning
Revert To Stone Age	Works. Its blind cones and horizon stay, its clutter does not

Cockpit layouts for PortalMFD are ready for these modded aircraft. Small screens that the table does not name (gauges, status strips, round dials) keep the original panels of the aircraft's author.

Aircraft	RWR tier	Screens	Notes
OA-27 Cavalier	1	One main screen (two halves)	
RAH-72 Knockout	1	One main screen (two halves)	

Aircraft	RWR tier	Screens	Notes
AH-31 Kukri	2	Middle of the wide screen (two halves)	Rank 1, its mast radar lifts it to Tier 2. The camera page opens on the right half
F-99 Shrike	2	Left, right, centre	
F-204 Icarus II	2	Left, right	Camera page on the left MFD
FQ-106 Kestrel	2	One main screen (two halves)	
MiG-21K Koshchei	2	Left glass, glass right of centre (upright)	The left page uses the full glass height
MiG-29 Fulcrum	2	One screen (two halves)	The analog cockpit of its plugin shows CNO-EVIL RWR data. No small attitude screen
F-16M King Viper	3	Left, right, centre	
KR-33 Agni	3	One main screen (two halves)	
MiG-29SMT	3	Top of the centre MFD (two halves)	The plugin MFDs stay, and its RWR shows CNO-EVIL bearings. No small attitude screen
Mirage 2000D	3	Left, centre scope, right (front seat)	Camera page on the centre scope. The rear seat sees the scope image. The plugin pages lie under the portal
MC-260 Chimera	3	Two MFDs at the pilot seat	The copilot screens copy them
FS-3 Ternion	3	Left (stores MFD), centre (radar MFD, `/`)	
F-22E Strike Raptor	3	Left, right, centre, lower centre	
FS-41 Eclipse	3	Two MFDs (tandem)	
PB-22 Trident	3	Outer (left) and inner (right) MFD at the pilot seat	Camera page on the outer MFD. The inner MFD has fewer pixels
Su-30VXN Gandiva	3	Two pages on the main glass: engine part (left), radar part (right)	The round dial, the HUD glass and the flight dials keep their vanilla look. No attitude display or datalink page

Text too soft on a small cockpit screen? `PortalMFDRenderScale` (1, 1.5 or 2, default 2) sets how sharp the pages are. Small slots get a larger texture by themselves.

37. Tuning the mod

The file is `BepInEx\config\com.cnoevil.nuclearoption.overhaul.cfg`. Close the game before you edit it. It has 297 keys in 18 sections, and each key has a description.

Section	Holds
01 General, 02 Debug Overlays	Master switch, debug keys, session log
03 Radar Physics	Threshold, notch, clutter, sea reflection
04 Electronic Warfare	Jammers, chaff, decoys, budget
05 Missile Guidance	Seeker activation, gimbal limits, ARM memory, watchdog
06 RWR Receiver	Tier values
07, 08, 09	Tracks, Q, CEP
10 AI Doctrine	Launch rules, tactics, radar silence against ARMs
11 Datalink	Network, warning gate
12, 13, 14, 15	HUD, map, TacScreen, thermal window
16 Countermeasure Programs	Keys and packages
17 Mission Placement	Dynamic placement of air defense
18 HMD Portal Pages	MFD copies on the HMD

Popular changes:

You want	Set
See where your missiles aim, for training	`ShowMissileAimpointOnMap` = true
See datalink missiles on the map	`EnableMapDatalinkMissiles` = true
See datalink strobes	`EnableDatalinkStrobes` = true
No jammer power limit	`EcmJamDuration_s` = 0
Bigger countermeasure load	`CountermeasureBudget`
Vanilla small flight screen	`CockpitAttitudeDisplay` = false
A different TacScreen	`TacScreenDisplayPreset`, `TacScreenThreatDisplay`
SAM sites where the mission file puts them	`DynamicPlacementEnabled` = false
The same SAM layout at every start	`DynamicPlacementSeed` = any number but 0
Sharper or cheaper cockpit screens	`PortalMFDRenderScale` (1, 1.5, 2; default 2)
MFD copies on the helmet	`EnableHMDPortalPages` = true

38. Troubleshooting

Problem	Likely cause	Fix
The mod seems off	Host without the mod, or another version	Check the log. On your own, press F10
`failed` is not 0 in the log	A game update changed something	Report the log
I cannot select an enemy radar	It is not located	Chapter 12
No notch cue	Tier 1, or the range to the threat is unknown	Chapter 6

Problem	Likely cause	Fix
Nothing warned me of the missile	No sensor saw it, or no collision course yet	Chapter 24
Chaff is not in the station list	A player in the session has no mod	Chapter 36
The thermal window does not open	No target camera, or not in the cockpit view	Chapter 9
The jammer does nothing	Empty supply	Chapter 14
`RCS` line missing	Gear not locked up	Chapter 5
My missile missed a marked target	Low Q, or an old track	Chapter 19
`LeftAlt` releases nothing	It was held over 0.5 s, or used with another key	Chapter 18
No selection box on the map	Q below 0.55	Chapter 28
A SAM went quiet after my ARM launch	It turned its radar off against the ARM	Chapter 22
Cockpit screen text is blurry	Low render scale	`PortalMFDRenderScale` = 2
Something odd happened		Record a session log (below)

Session log. For a bug report, set `DebugLogging` to true. The mod then writes one file per game session to `BepInEx\CNO-EVIL-Logs` (`session-<date>-<time>.log`), with the mod and game versions, your other plugins and every setting you changed. Press `F6` at the moment something odd happens: it writes a numbered mark and a snapshot of you, your targets, missiles and the radars on you, and the screen shows `LOG MARK n`. Send the newest file with the mark number. The `DebugLog...` switches in section 02 choose the categories (radar, track, missile, AI, EW, net, placement, data; all on), and `DebugLogKeepSessions` (10) sets how many files stay. Errors and warnings still go to `BepInEx\LogOutput.log`.

39. Glossary

Term	Meaning
ARH	Active radar homing: the missile has its own radar
ARM	Anti-radiation missile: homes on a radar signal
Basket	The area around the datum where a seeker finds the target
Bearing	Direction from a receiver to an emitter
Burn-through	The point where the echo beats the jamming
CEP	Radius that holds half of the weapon offsets
Clutter	Echo from the ground or the sea
Datum	The point a weapon flies to before its seeker finds the target
DRFM	The active decoy: a repeater in a flare cartridge
ECM	Jamming
EMCON	Emission control: a radar goes silent on purpose, for example against an ARM
ELINT	The Tier 4 receiver of an aircraft with a radome

Term	Meaning
ESM	RWR reception of a radar too weak or too far to detect you. Gives a bearing, no warning
Fix	A position calculated from bearings
HERM	Echo of helicopter rotor blades
HOJ	Home on jam
Illuminator	The radar that lights the target for a SARH missile
IRST	Passive bearings from the thermal window
Located	Position good enough for the map
MAWS	Own missile sensor, 5 km with line of sight
Memory coast	A seeker keeps its lock for a moment after the notch rejects the target
Multipath	The echo off the sea or ground that adds to or cancels the direct echo
Notch	The band of radial speeds a radar removes with the clutter
Pitbull	The moment an ARH seeker starts to transmit
Q	Track quality, 0 to 1
RCS	Radar cross section
SARH	Semi-active radar homing: the missile hears the echo of another radar
SNR	Signal-to-noise ratio
Spotter	Any unit with optics
Strobe	A line or cone that shows an RWR bearing
Track	What a faction knows about one target

40. Quick reference card

Situation	Action
Strobe, no position	Fly across the bearing
`ID LOW` / `ID MED`	Stop transmitting. Change altitude
SNR pip near the tick	Descend, notch or use terrain
`[ LOCK ]`	Prepare to notch
Low alarm, cyan `M`	Beam now, descend, decoy
`[ MISSILE ]`	Beam the guiding radar, descend
`[ PITBULL ]`	Beam, `LeftAlt`, turn hard, descend
`[ HOJ RISK ]`	Notch or jammer off
`[ BURN-THROUGH ]`	Jammer off, notch, terrain
`[ NO CLUTTER ]`	Descend
`ID OPT`	Terrain or distance
Pale white marker	Do not fire yet
Amber marker	Seeker weapons
Red marker	All weapons
Battery low	Stop jamming, 30 s to full
SAM silent after your ARM	It waits. Stay out of its zone
Something odd happened	`F6` (with `DebugLogging` on)